

To What Extent is the Royal Air Force Still Fit for Purpose?



Sam Travaglia

Bishop of Llandaff CiW High School

Advanced Skills Baccalaureate Wales Qualification- Individual Project

16 January 2025

Contents

Rationale	3
Aims & Objectives	4
Research Plan	5
Data Analysis	10
Conclusion	42
Bibliography	44
Appendix	57

Rationale

To determine whether the Royal Air Force is still capable of projecting power, protecting strategic interests, upskilling, fairly treating, and protecting its personnel, and is sufficiently prepared for any sort of local, regional, or global conflict.

The Royal Air Force is the UK military's Air and Space defence arm, with goals of securing and protecting UK airspace, protecting UK assets, providing humanitarian assistance, and defending British interests both at home and abroad. Having existed continuously since 1918, it is the world's oldest air force, and has taken part in many conflicts, operations, and objectives since its inception, including WW2, the Falklands War, and Operation Telic (Invasion of Iraq) to name but a few.

However, it has recently come under significant criticism for major cuts to fleet size, personnel numbers, and suffering a significant number of personnel scandals, including the Recruitment scandal and the Red Arrows scandal. This has negatively impacted the public's overall view of the Royal Air Force, calling into question its leadership, culture, and capability, and arguably bringing the entire service into disrepute. Furthermore, it now severely lacks behind countries like China in terms of overall size, bringing into question its ability to complete strategic objectives, or even fight a war if needs be.

The purpose of the Project is to analyse whether the RAF can sufficiently meet these goals.

Aims & Objectives

Aim 1: To analyse overall trends in the fleet size of the RAF, and the reasons for these trends

Objective 1: To examine the overall changes of the size of the RAF through time

Objective 2: To discuss specific reductions in fleet size

Objective 3: To understand the causes of the RAF's overall fleet size adjustment

Aim 2: To compare the current size of the RAF to the sizes of other air forces, and determine whether the RAF is able to fulfil its current requirements

Objective 1: To compare the current size of the RAF to that of other air forces

Objective 2: To discuss the recent personnel controversies and their public effects

Objective 3: To determine the current requirements of the RAF and its current capabilities

Aim 3: To determine the current overall effectiveness of the RAF and how this may change in future

Objective 1: To analyse the current effectiveness of the RAF (including recent operational involvements)

Objective 2: To discuss the future of the RAF

Objective 3: To determine ways in which the RAF could revive its public image

Research Plan

Aim 1: To analyse overall trends in the fleet size of the RAF, and the reasons for these trends

Objective 1: Examine the overall changes of the size of the RAF through time

Objective 2: Discuss specific reductions in fleet size

To successfully meet the objectives, I will need to obtain accurate quantitative and qualitative data about how the size of the Royal Air Force has changed over its history, addressing both equipment and personnel numbers. This is important as the size of any military branch usually provides a good indication on its overall ability to wage warfare effectively and project power in a local, regional, or global setting. Additionally, I will need to analyse specific reductions in fleet size by categories such as aircraft type, since a reduction in a type of aircraft can have critical strategic impacts on the ability of the Force to project power and meet operational objectives. Obtaining this data will also help me with answering Aim 2.

One method I will use to collect this data is via the use of news articles. This is due to their dissemination of timely information as well as the relatively high likelihood of them including visual mediums to convey information (e.g. graphs). Additionally, I will use online research papers, such as the House of Commons Defence Committee's 2023 paper "*Aviation Procurement: Winging It?*", due to their high likelihood of accuracy and ability to discover other sources via citations, as well as relevant government sources due to their wide timespan and ability to cross-reference. Using all these data sources in conjunction will allow me to ensure that the results I obtain for these objectives are as accurate as possible, by comparison between sources, and exclusion of any sources likely to be inaccurate/outdated/otherwise misleading.

I will find useful and relevant sources through the use of online tools, such as Google Scholar. This will involve searching via keywords (e.g. "RAF fleet size," "RAF personnel numbers"), filtering results by timespan (e.g. 1991-2024), and selectively including/excluding domains based upon whether they can provide trusted information (i.e. staying clear of tabloid articles unless they are a primary source of relevant information). I will also endeavour to use government information and resources, such as the House of Commons Library, to search for and obtain relevant documentation and other information. Once this data has obtained, I will transfer it into a suitable visual medium, such as graphs and charts, to best highlight the change in overall fleet size and specific fleet size reductions. This will be done using Microsoft Excel, due to my familiarity with its User Interface and relative ease to use. I will aim to create a number of graphs, such as line graphs, pie charts, and bar charts, as these can best convey change in overall numbers as a product of time, and composition of aircraft type. I will also use line and bar graphs to display any changes in personnel numbers over time, to best display any trends in these figures.

Objective 3: Understand the causes of the RAF's overall fleet size adjustment

In order to successfully meet the objective, I will need to obtain accurate quantitative and qualitative data about the causes of the changes in the overall fleet size of the RAF. This is essential as it will allow me to link what factors affect the size, and therefore the capability of the air force, allowing me to determine which factors may be most significant in causing this change in size, as well as potentially considering ways in which any negative trends in fleet size could be reversed, which will assist with answering Aim 3.

One method I will employ to collect this data is the use of online government sources and documents, due to their widespread public availability and inclusion of historical data. This will be used in conjunction with news articles, from sites like UK Defence Journal, due to their timely dissemination of relevant information, and interviews, as they allow for spontaneously generated questions to be asked and provide an opportunity to

collect relevant and expert qualitative insight. Using all of these sources in conjunction will allow for a wide array of qualitative and quantitative data to be obtained, allowing for links to be drawn between factors, and potentially allowing me to suggest the most significant factor that has led to changes in the RAF's fleet size and personnel numbers.

Similarly to the previous objectives, I will find useful and relevant sources using online research tools, such as Google Scholar. This will involve searching using keywords (e.g. "RAF annual budget," "RAF operational deployments"), filtering results by timespan (e.g. 1991-2024), and selectively including/excluding domains based upon whether they can provide trusted information (i.e. staying clear of tabloid articles unless they are a primary source of relevant information). I will also endeavour to use government information and resources, such as the House of Commons Library, to search for and obtain relevant documentation and other information, such as budgets and Government policy. I will also produce a list of questions (e.g. What do you think are the most significant factors causing changes in the RAF's fleet size/personnel numbers?) to ask a former member of the British armed forces, to obtain expert insight into what effect these spending patterns are having on the capability of the British armed forces.

Once the relevant information has been collated, I will aim to represent this data by creating visual mediums to convey the obtained quantitative information, such as line graphs to display changes in funding over time, as well as using word clouds to display obtained qualitative data, to convey the occurrence of key words in both the analysed sources and the transcripts of the interviews I conduct. This will allow for a clear method of conveying the importance of certain factors, since the frequency of occurrence of a key word is likely to correlate to its relative importance with regards to this objective.

Aim 2: To compare the current size of the RAF to the sizes of other air forces, and determine if this size is appropriate and sufficient to allow it to fulfil its current operational requirements

Objective 1: Compare the current size of the RAF to that of other air forces

To successfully meet the objective, I will need to obtain accurate quantitative and qualitative data about the size of other air forces, both from countries considered allies (e.g. USA, France, Germany) and potential adversaries (e.g. China, Russia). This will allow for a clear comparison of the likely capability of other air forces, and, by comparing the strength of each air force with its overall global power, it can be determined whether the overall change in the RAF's fleet size is an isolated incident or can in fact be attributed to a wider regional, or even global, trend.

Similarly to Aim 1, I will employ the use of news articles, due to their dissemination of timely information as well as the relatively high likelihood of them including visual mediums to convey information (e.g. graphs), to be supplemented via the use of online research papers due to their high likelihood of accuracy and ability to discover other sources via citations, as well as relevant government sources due to their wide timespan and ability to cross-reference, as well as data from multinational organisations, such as NATO.

As in Aim 1, I will find useful and relevant sources through the use of online tools, such as Google Scholar. This will involve searching via keywords (e.g. "USAF fleet size," "USAF personnel numbers"), filtering results by timespan (e.g. 1991-2024), and selectively including/excluding domains based upon whether they can provide trusted information (i.e. staying clear of tabloid articles unless they are a primary source of relevant information). I will also endeavour to use government information and resources, such as budgetary information from NATO, to find accurate and relevant information. I will aim to include as much financial information as possible, since this will give the best indication of not only the overall size of a country's air

force, but also how importantly it is perceived to the country, if military expenditure is taken as a percentage of GDP. Once this data has obtained, I will transfer it into a suitable visual medium, such as graphs and charts, to best compare the obtained data to that of the RAF. This will be done using Microsoft Excel, due to my familiarity with its User Interface and relative ease to use. I will aim to create a number of graphs, such as line graphs, pie charts, and bar charts, as these can best convey change in overall numbers as a product of time, and composition of aircraft type. I will also use line and bar graphs to display any changes in personnel numbers over time, to best display any trends in these figures.

Objective 2: To discuss the recent personnel controversies and their public effects

In order to answer this objective, I will need to obtain sufficient examples of recent personnel controversies and collate substantial qualitative data about the sway in public opinion in reaction to these revelations. This will allow me to clearly determine the severity of these controversies in the public eye and see how they have affected the public's overall belief in the RAF.

In order to collect relevant and useful data, I will primarily use public surveys to answer this objective, due to their ability to reach large audience quickly, as well as obtained data being able to be analysed for trends. This data will be supplemented with the use of news pieces, due to their serious and straightforward handling of serious fact, such as the BBC's 2023 article into the Red Arrows Scandal, as well as interviews, to collect expert analysis.

To collect this data using the above methods, I will create a survey on Microsoft forms, which will include questions such as *"What is your current opinion of the RAF?"*, to gauge current overall viewpoints of the force, as well as questions such as *"In 2023, the RAF was found to have unlawfully discriminated against white males to boost its diversity. How familiar are you with this?"*, to gauge how much public opinion has shifted after hearing of these controversies. The survey will be sent out to people in Years 12 and 13, using the *Intake18* and *Intake19* email addresses. I will also exclude any responses that could be considered irrelevant or inappropriate, to prevent an inaccurate skew of data, although I will take every opportunity to avoid this, to prevent excluding useful data. I will also emphasize the possibility of the Hawthorne bias in my data analysis, as it is entirely possible a person undertaking a survey will alter their responses, although the chance of this should be minimised provided the survey is set to anonymous.

Once this data is obtained, I will compare it to the answers received from an interview with a former member of the British armed forces, in order to determine whether public opinion is in line with that of the opinions of ex-servicemembers. This will help to determine whether the general public's opinion is based on similar data, as any disagreement in overall opinion on the RAF may be as a result of one group being mis-informed or holding internal biases towards or against the RAF and British armed forces as a whole. I will also undertake statistical analysis on some questions, in order to fully understand how much hearing about the recent RAF scandals has affected public opinion.

Objective 3: To determine the current requirements of the RAF

To answer this objective successfully, I will need to perform an in-depth analysis of the current roles of the RAF, including its battle doctrine, role inside of NATO, and its involvements in the cyberspace and space domains. This is essential to answering the objective as it will ensure that all aspects of the RAF's strength are being fully analysed, not just its traditional 'fighting force' of solely aircraft. Since the RAF is now conducting warfare from many angles, not just via powered aircraft through the domain of air, but now the cyberspace and space

domains in addition, as well as being a part of NATO, and following the 'whole force' concept, where every part of the service is considered essential to its smooth running, all aspects of the RAF's capability, expertise, and doctrine must be fully taken into account.

One method I will employ to collect this data is the use of online government sources and documents such as Joint Doctrine Publication 0-30, the UK's keystone document on Air Power Doctrine, due to their widespread public availability and inclusions of current strategic decisions. This will be used in conjunction with news articles, due to their timely dissemination of relevant information, and interviews, as they allow for spontaneously generated questions to be asked and provide an opportunity to collect relevant and expert qualitative insight. Using all of these sources in conjunction will allow for an accurate picture of the RAF's current shifts in resource allocation, and how its operational doctrines are shifting. I will also use news articles to determine whether the RAF urgently requires more independence from allied nations, such as the USA, due to recent political events, as outlined by Laura Kuenssberg's 2024 BBC article into the re-election of Donald Trump.

Once this data has been obtained, I will use it to explain any shifts in the RAF's methods of controlling the air domain, as well as analysing whether they have begun to shift into controlling various other domains, such as the space domain. It will also be used to determine whether the RAF is promoting technological advancement over sheer numbers, and whether this approach is more beneficial or detrimental to the RAF.

Aim 3: To determine the current overall effectiveness of the RAF and how this may change in future

Objective 1: To analyse the current effectiveness of the RAF (including recent operational involvements)

In order to successfully answer this objective, I will need to collate sufficient quantitative and qualitative data on the current roles and responsibilities of the RAF. This will include recent operational involvements (e.g. Operation Shader/Intervention against Dā'ish (ISIS)), as well as its current ongoing roles within and outside of NATO, including domestic and foreign air policing missions, humanitarian efforts, and training of foreign militaries. This will determine whether the RAF is still capable of sufficiently completing several operational objectives simultaneously, which would be a strong indication of its effectiveness and overall capability if so.

One method I will employ to collect this data is the use of online government sources and documents, such as NATO official press reports, due to their widespread public availability as well as their in-depth insight into complex and shifting topics. This will be used in conjunction with news articles, due to their timely dissemination of relevant information, and interviews, as they allow for spontaneously generated questions to be asked and provide an opportunity to collect relevant and expert qualitative insight. These sources will allow for analysis into the RAF's current and past deployments and their relative success.

Once sufficient data from the above sources has been obtained, I will use it to determine the areas in which the RAF has had a presence in over the past year, which will be plotted onto a map for ease of understanding and comprehension for the reader. Additionally, I will analyse both Operation Pitting (Kabul airlift) and Operation Azotize (Baltic Air Policing) in depth in order to provide a detailed judgement on the current operational abilities of the RAF, their strengths and weaknesses, and whether the previously discussed items (e.g. fleet size change) are responsible for these strengths and weaknesses.

Objective 2: To discuss the future of the RAF

To answer this objective, I will need to collate qualitative data about the future of the RAF. Specifically, I will aim to understand what aircraft the RAF is investing in, where resources are currently being diverted to/funnelled from, and how the overall culture of the force is changing. This will help me to understand how the RAF is currently changing, and how it is likely to change in future. Ultimately, the path the RAF chooses to take today will greatly influence both its current and future ability to meet its operational objectives and is therefore essential to being understood to fully answer the overall question of the project.

One method I will employ to collect this data is the use of online government sources and documents, such as reports from HM Government of the future of defence spending, due to their widespread public availability as well as containing public policy relevant to the future of the RAF. This will be used in conjunction with news articles, due to their timely dissemination of relevant information. These sources will allow me to generate insight into how the RAF will change in future.

I will use the data to understand important factors of the future success of the RAF in future, such as changes to fleet, budget, and leadership. Using this data, I will explain how the fleet of the RAF is planned to change in future, describing any new aircraft planned to enter service, explaining the benefits of these aircraft, and analysing how it will change the strength of the RAF. Additionally, using budgetary information will clearly show the ability of the RAF to procure new equipment and support more operational involvements, with this budgetary data being presented with a line graph. Furthermore, changes in leadership will be explained, using analysis from the news and public statements to determine the importance of these changes and the shift in attitudes from those at the top of the RAF.

Objective 3: To determine ways that the RAF could revive its public image

In order to answer the objective, I will need to obtain accurate and sufficient qualitative data about the best methods the RAF could employ to successfully revive its image. Since the RAF is a government funded organisation, a drop in overall public support is likely to reduce the political will to keep the RAF's funding plentiful, especially with an ever-increasing strain on other, more public-facing services (such as the NHS). As such, it is essential that I can find measures that the RAF could take to revive its image, as these will best indicate whether the RAF truly wants to revive its image by taking these measures or not.

One method I will use to answer this objective is using opinionated news pieces, due to their insight into relevant topics and can stimulate discussion and debate, as well as the possibility of them being written by a subject matter expert. I will also use public surveys, to collate relevant public insight, giving me plentiful data to analyse, as well as hosting an interview with a subject matter expert, in order to acquire thoughtful and in-depth answers. I will use this information to determine important values such as the amount of time the public estimate it will take for them to regain trust in the force, and the most effective ways that the RAF could reform itself to regain public trust most efficiently.

Combining the results from the above methods, I will analyse which possible methods would be most favoured by both the public and industry experts, with these methods being obtained from the analysed opinion pieces. I would also generate an estimation of how long the general public and subject matter experts think would take the RAF to successfully revive its public image. I would also question the respondents to my survey the likelihood that they would join the RAF and use this qualitative data behind their reasoning to determine any other issues with the RAF, of the factors encouraging people to join the force, whilst being careful to factor in any biases from the respondents.

Data Analysis

Aim 1 – To Analyse Overall Trends in the Fleet Size of the RAF, and the Reasons for These Trends

Undoubtedly, any organisation, military or civilian, past or present, can rely on sheer size to achieve its goals. Provided enough resources, human, technological, financial, any obstacle can be cracked, no matter the time, no matter the size. Any military branch, including the Royal Air Force, is no exception to this rule. Aircraft are fierce yet fragile, dangerous but delicate. Despite their ability to be anywhere in the world, at moment's notice, with the power to start or end a conflict strapped to the underside of their wings, it doesn't take much to knock them out of the sky. Therefore, by holding an advantage of scale, any air force is automatically at an advantage, being able to replace lost equipment more readily, and being able to deploy more easily and readily, ultimately increasing its global influence and power.

However, a basic analysis of the RAF's overall change in fleet size since the 1990s, the subject of Objective 1, has found a drastic drop in fixed-wing aircraft numbers over the past 30 years, indicative of an overall drop in total aircraft numbers since the end of the Cold War. According to the House of Commons Defence Committee white paper *'Aviation Procurement: Winging It?'*, the RAF has suffered a 64.6% decrease in fixed wing aircraft numbers since 1990, from 579 to 205 (House of Commons Defence Committee (2023)), with UK Defence Journal claiming a 22% size reduction from 724 in 2006 to 564 in 2023 (George Allison (2024)).

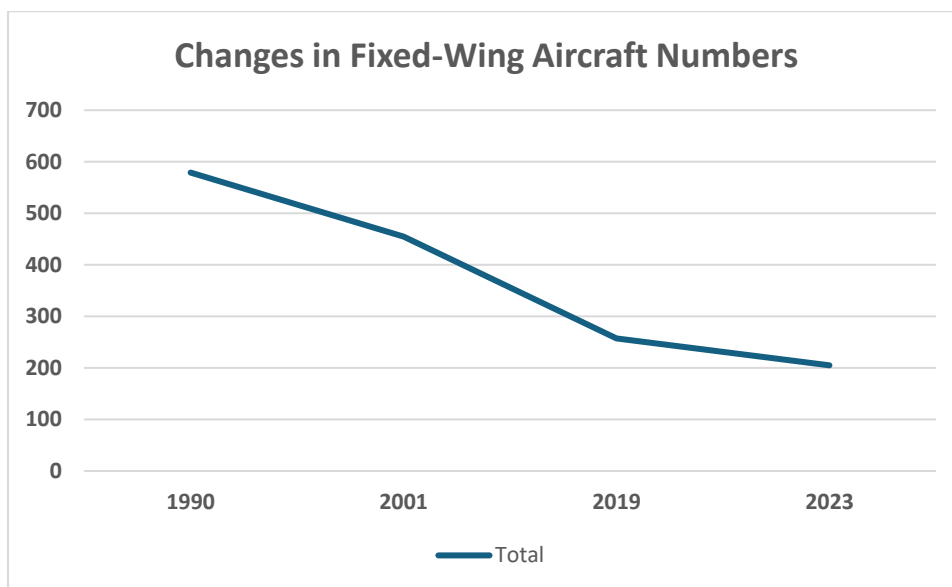


Figure 1- A graph of the total number of fixed-wing aircraft in the RAF, by year (House of Commons Defence Committee (2023))

Perhaps more concerning for the force, a Freedom of Information request answered by the Defence Statistics branch of the Ministry of Defence (MOD) found that while overall regular numbers across all British armed forces have dropped by 50.7% since 1990 until 2016, RAF personnel numbers have dropped by a staggering

63.3% in the same period, their lowest numbers since 1935, with current numbers now at 32,228 as of 2024, according to the House of Commons Library (Defence Statistics (2017)) (Esme Kirk-Wade, 2024). This means that for the first time, all 3 services are below their respective target sizes, with the RAF currently doing worst at 10% below target. Furthermore, the RAF has practically no operational reserve, with only 1432 people in full-time reservist service, meaning that in the event of a major conflict, where the entire force is needed, the RAF will have access to barely a handful more troops. This is especially staggering, considering that according to The Times, the number of civil servants working in the Ministry of Defence not only outnumbers the number of personnel in the entire RAF, but in the Royal Navy, Royal Marines, and RAF *combined* (63,702 vs 57,260) (Larisa Brown (2024)).

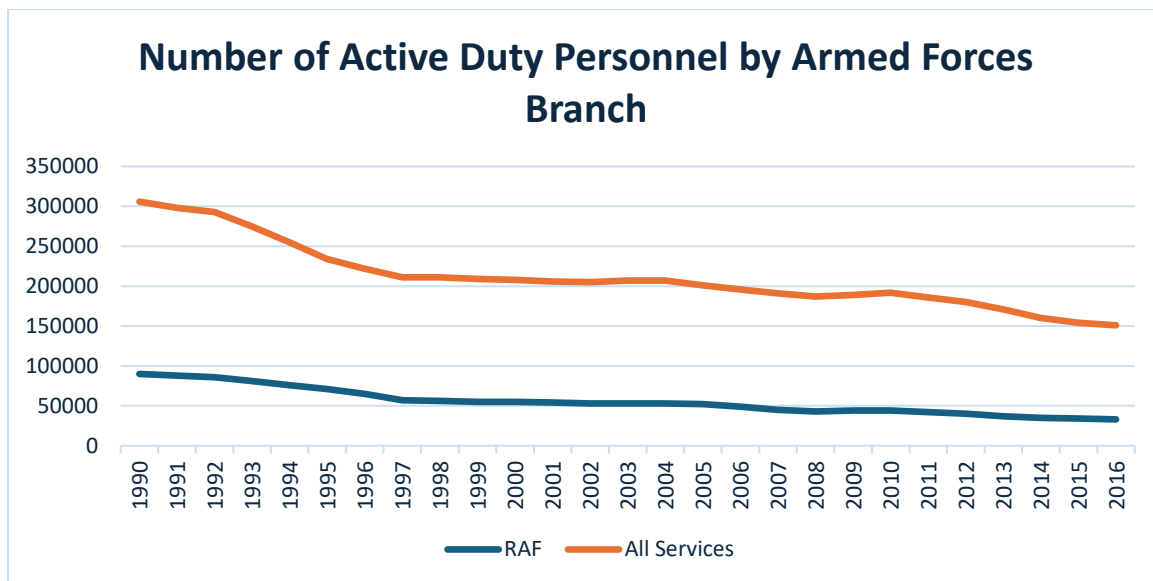


Figure 2- A graph of the total number of active-duty personnel in the RAF and the armed forces, by year (Defence Statistics (2017))

It should of course be said that size is by no means a guarantee of success in a battlefield, or rather any sort of environment. As said by the 2023 House of Commons Defence Committee paper, *Aviation Procurement: Winging it?*, “It is true that mass alone does not win wars.” However, it continues “There is, however, a fine balance to be struck between exquisite capability and aircraft numbers. Several witnesses expressed concern that the size of the current combat air fleet meant it would not be able to endure the levels of attrition which could occur were we to go to war—a scenario which until recently seemed remote, but is now a very real possibility.” (House of Commons Defence Committee (2023)). Furthermore, Joint Doctrine Publication 0-30, the UK’s keystone air domain doctrine publication, an up-to-date, unbiased source, states that “The UK maintains highly credible air power but cannot provide the full breadth of air power capabilities and enablers to generate, coordinate and sustain the mass required to conduct high-intensity operations alone.” (Ministry of Defence (2022)).

Clearly, the RAF has experienced a very substantial drop in overall size, both in terms of personnel numbers and number of aircraft. This has led to speculation, both by official sources and independent analyses, that the RAF would struggle, and may even face the possibility, of losing a major conflict should one occur.

Of course, any strong force doesn’t just need sheer numbers, but relevant and useful resources. Just like how a team of 11 goalkeepers probably wouldn’t be successful in the Premier League, likewise any air force needs to have certain aircraft types that are specialised in one of, or several of, 4 basic operational capabilities. Broadly, these capabilities, as stated in JDP 0-30, fall under the categories of Control of the Air, Attack, Air Mobility, and

Intelligence, Surveillance, and Reconnaissance (ISR). These operational categories are all inter-dependent, with the success of one requiring on the success of all. For example, it is impossible to conduct a ground strike mission on an enemy base (Attack), with no capability to transport munitions to the relevant forward operating bases (Air Mobility). Therefore, for an air force to truly be capable and an influential force, it must be able to meet operational requirements for all 4 of these mission types.

However, because of the RAF's overall decline in fleet size, it has also lost a substantial amount of fleet size for each air capability, with some more adversely affected than others.

One of the main areas the RAF is severely lacking in is its ISR fleet, specifically in its Airborne Early Warning & Control (AEW&C) aircraft. These aircraft patrol the sky, controlling the battlefield and observing for any threats in the nearby operational battlespace, keeping allied aircraft safer and making them a true force multiplier. Up until recently, the RAF's sole AEW&C aircraft was the Boeing E3-D Sentry, of which it held a fleet of 6 in the 2010s. However, from 2018 until 2022, the Sentry fleet was retired in favour of obtaining a more technologically advanced Airborne Warning And Control System (AWACS), namely the Boeing E-7 Wedgetail. Although this decision was made to modernise the RAF's ISR fleet, in a rather counter-intuitive manner, the RAF retired its entire Sentry fleet before receiving any E-7s, and is still yet to receive any.

It has now been over 2 years since the RAF retired its entire AWACS fleet, and it is still yet to receive any aircraft that can provide any sort of AEW&C capability at all. As such, the RAF is missing an extremely important piece of capability. According to the Royal Aeronautical Society, *"AEW&C has never been as important as it is today, providing offensive and defensive roles to combat aircraft over Syria, providing 'big picture' situational awareness to coalition aircraft in the battle space, as well as deconflicting the airspace, providing updates on the movements of non-coalition aircraft. The importance of AEW&C platforms has once again been highlighted to air arms as a vital component of airspace management, long-range detection and control in the battlespace."* (Khalem Chapman (2018)). As a result, the RAF will likely suffer in any future aerial conflicts, due to the loss of the influential and powerful AWACS systems it once had in its arsenal.

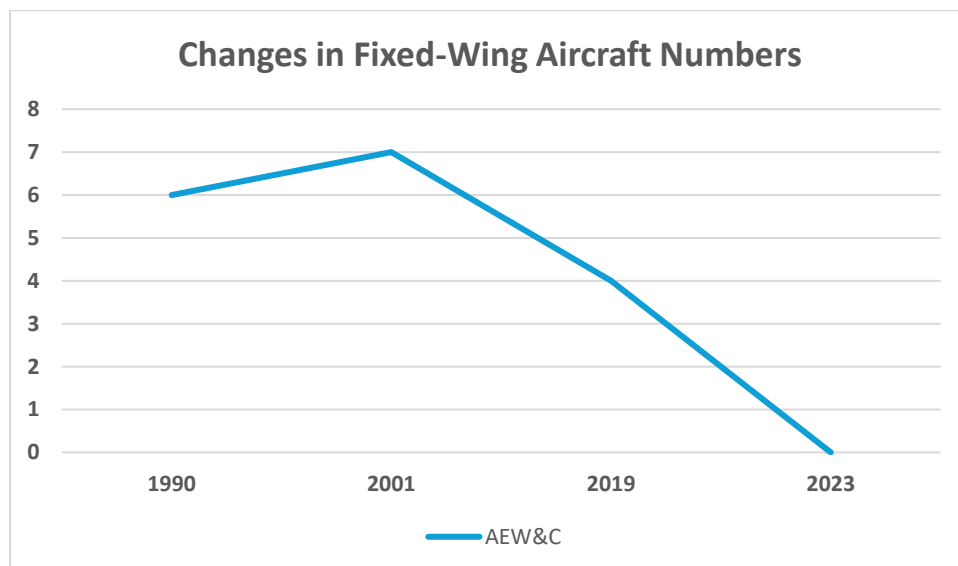


Figure 3- A graph of the total number of Airborne Early Warning & Control aircraft in the RAF, by year (House of Commons Defence Committee (2023))

Furthermore, the RAF came under substantial criticism last year for retiring its entire C130J Hercules fleet, one of its longest-serving, useful, and reliable aircraft, described by an RAF source as the *"backbone"* of the UK's

transport fleet, according to Sky News (Deborah Haynes (2023)). As a result of the 2010 Strategic Defence and Security Review, it was determined that holding on to the Hercules fleet beyond the early 2020's would be financially unviable, due to the excess wear sustained in the Afghanistan and Iraq Wars (Lima Charlie (2023)). Despite this, real attempt has been made to replace this substantial loss of airframes, with plans to increase the size of the Atlas C-1 fleet cancelled due to further budgetary constraints. This has resulted in huge strain on the RAF's air mobility fleet, as now there is increased workload on a dwindling number of transport aircraft, even more so considering that, according to Forces News, *"roughly one third of the fleet is fully available for tasking and up to another third in maintenance [...] with only 22 aircraft in service, the RAF will be lucky to have between 12 to 15 available for regular use."* (Lima Charlie (2023)).

Moreover, not only is the RAF suffering from a lack of Air Mobility aircraft, but the aircraft themselves are less capable than the C130s. Crucially, the RAF's Atlas fleet is not certified for many special forces-specific missions, such as paratroops and boat drops, allowing for surgical deployment of highly specialised and capable troops, such as the Special Air Service or Special Boat Service (Deborah Haynes (2023)). As such, the UK Armed Forces now lack the ability to insert special forces vehicles inside hostile territory. Perhaps most concerning, the RAF's Atlas fleet, as well as the entire A400 fleet across all nations that operate it, is issues with corrosion of essential components, like the landing gear, hydraulics and electronic coupling rings, clamps, and screws, as well as gearbox issues, putting both the engines and the entire aircraft out of service as it awaits maintenance (Tim Martin (2023)). Although this issue is now on the way to being solved, it has cost Airbus, the aircraft's manufacturer, €1.2 billion in fines, and has led to yet more negative press of the RAF, with Atlas aircraft on the ground at RAF Brize Norton with their engines removed for maintenance now a regular sight (John Hill (2024)).

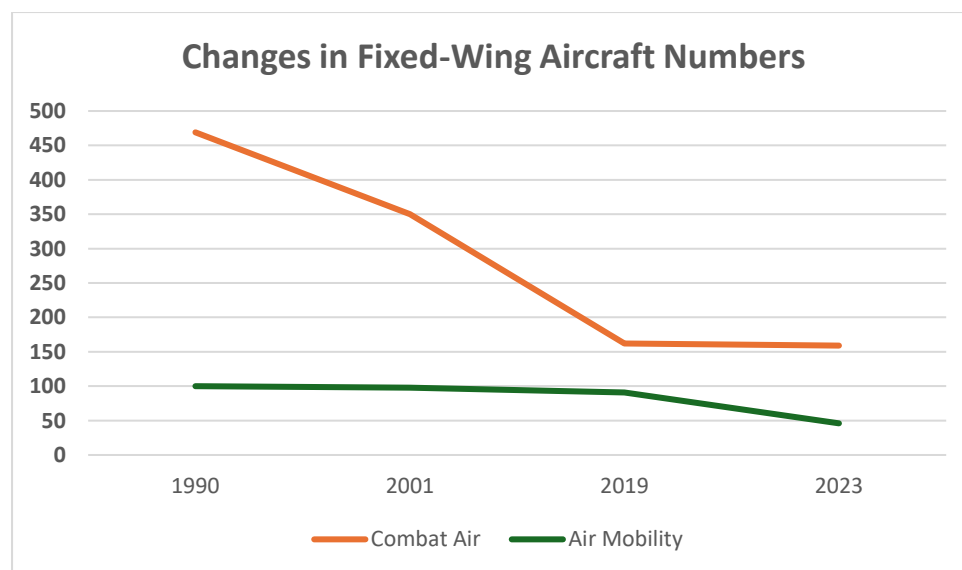


Figure 4- A graph of the total number of combat and air mobility aircraft in the RAF, by year (House of Commons Defence Committee (2023))

In addition, the RAF has faced a substantial dip in numbers in its fast-jet fleet, reflecting a reduction in capability both in terms of its capability for Control of the Air and Attack. This is due to the RAF's decision to invest into more capable and advanced multirole stealth fighters, namely the F-35B. This allows for the RAF to rely on less airframe variants to complete various missions, simplifying the training, procurement, and maintenance processes. These aircraft can be used to fulfil a variety of mission requirements, such as Suppression of Enemy Air Defences (SEAD), Aerial Bombing, Reconnaissance, and Air Combat. Although this allows the aircraft to be more cost-effective, the F-35, particularly the -B variant, being extremely expensive, with price estimates at around \$121.8 million *per airframe*, compared to \$59.6 million per Panavia Tornado aircraft (the aircraft they replaced), and \$105.7 million per Eurofighter Typhoon FGR4 (the UK's other main Fast Jet aircraft). And, although these F-35s are carrier-capable, both the RAF and Royal Navy do not possess a sufficient number of aircraft to completely fill both Queen-Elizabeth class aircraft carriers. This is before considering the various technical issues the carrier fleet have fallen victim to over the past 5 years, with

propeller shaft issues plaguing HMS Queen Elizabeth in both 2019 and 2024, which prevented her deployment to a major NATO exercise.

These shortages in aircraft numbers has led to backlogs in the training pipeline for pilots in the RAF, where in 2019, according to the House of Commons Defence Committee, *“training of the RAF’s fast-jet pilots took an average of 7.1 years, compared with the Department’s optimum time of 3.9 years”*, with these issues being attributed to *“several underlying causes [...], including a shortage of qualified instructors and poor aircraft availability”* (House of Commons Defence Committee (2023)).

Clearly, the RAF has noticeable shortfalls in most areas of its aircraft fleet. This has compromised its ability to operate in missions in all 4 tenants of air power, hugely reducing its ability to project power across the globe, or even defend its territory in a large-scale conflict, especially in a war of attrition-type scenario.

Although the MOD’s budget currently stands at £45.9BN, the overall MOD budget has stagnated at 2% of GDP since 1998. With this stagnation in overall funding, some sources estimate deficits of 10,000 personnel and £2bn. This has caused that RAF to take a number of unpopular measures to reduce budgetary shortcomings, including the controversial decision to scrap the Hercules fleet, which was due to substantial budget shortcomings. Furthermore, according to Mr A Ashment, an army reservist I interviewed, military personnel are being *“sent into operational zones now, with limited resources”*. He continued by saying *“certain senior leaders within any of the forces are under [financial struggle], and for that reason, they’ve had to cut regiments, fleets, aircrafts, whether it’s sky, land sea, warfare”*, describing the overall situation as *“a struggle”*.

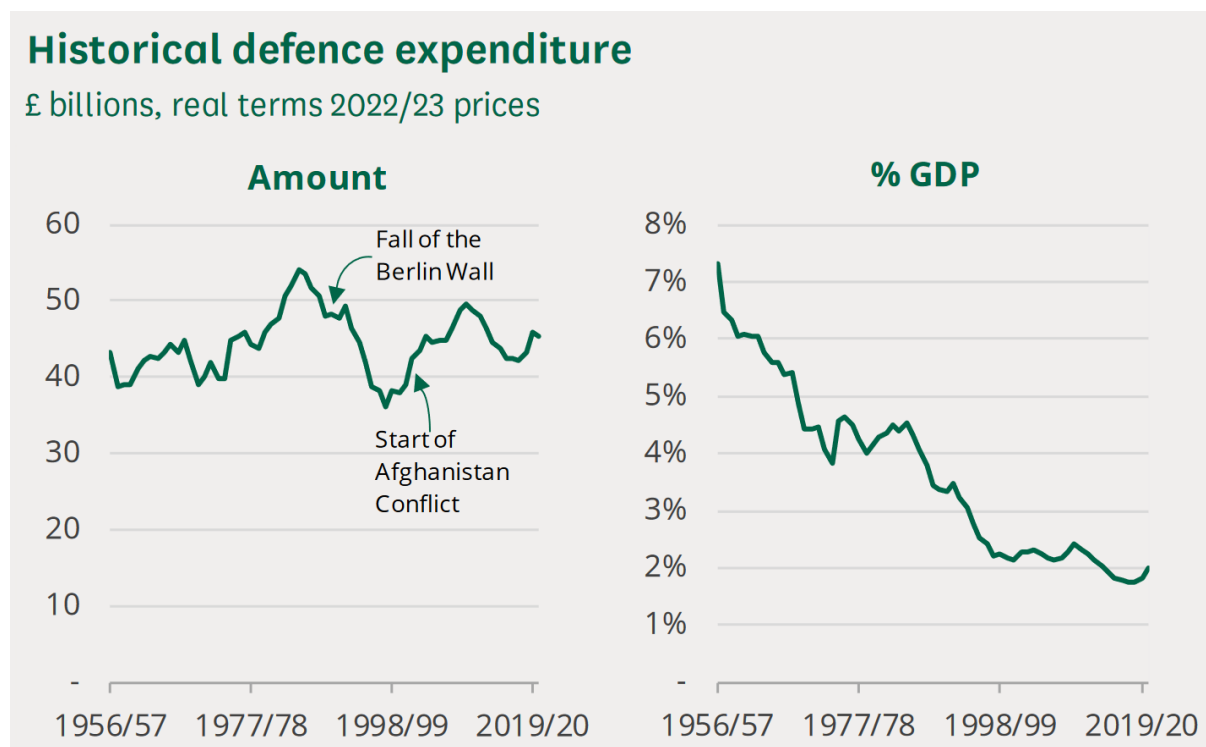


Figure 5- A graph of total government defence expenditure, by year (Esme Kirk-Wade (2024))

In the era of austerity, the public purse is lighter than ever. As described by Sky News, the vast majority of decisions taken by the RAF resulting in gaps in capability have been as a result of budget cuts. These decisions include, but are not limited to, the axing of the C-130 fleet, the reduction of the E-7 order from 5 airframes to 3, the delays in flight training for new recruits, sometimes being put on hold for months whilst waiting for opportunities to progress to flight training, the role they signed up for (Deborah Haynes (2023)) This recent

source, using data from the Government, clearly and reliably shows that these budget shortcomings are having severe effects on the RAF. These budget shortcomings are now resulting in public uncertainty about what could be cut, including the Red Arrows, as erroneously suggested by the Daily Mail (Jon Lake (2025)).

Aim 2- To Compare the Current Size of the RAF to the Sizes of Other Air Forces, and Determine if This Size Is Appropriate and Sufficient To Allow It To Fulfil Its Current Operational Requirements

Undoubtedly, the RAF’s fleet size has been declining substantially since the end of the Cold War, resulting in concern and scrutiny from both the press and Parliamentary reports. However, very little press has been given to the changes in the air forces of other nations, with the exception of China and Russia, believed to be the most likely adversaries to the RAF in the event of a conflict. Therefore, any change in the RAF’s fleet size should be compared to changes in the size of similarly sized air forces, in order to truly determine whether these trends are unique to the RAF, or as a result of a wider trend. If one is to consider the total aircraft a nation is in possession of, using the US, Italy, France, and Germany as our comparisons, we observe the following:

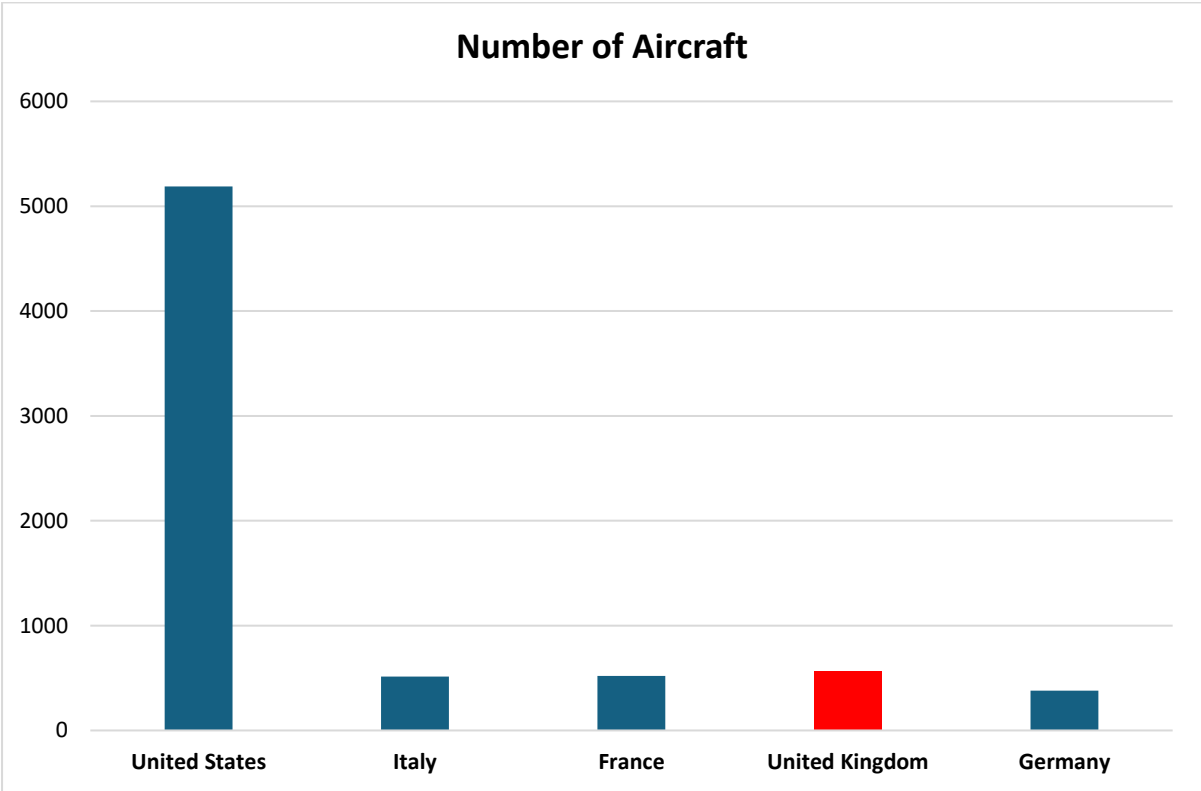


Figure 6- A graph of the total number of aircraft in service in certain countries (World Directory of Modern Military Aircraft (WDMMA) (2025))

Of course, figure 6 seems to show that the RAF is substantially behind its (arguably) greatest ally, the United States. However, by removing the US from the above graph, we see a far less bleak picture emerges for the RAF:

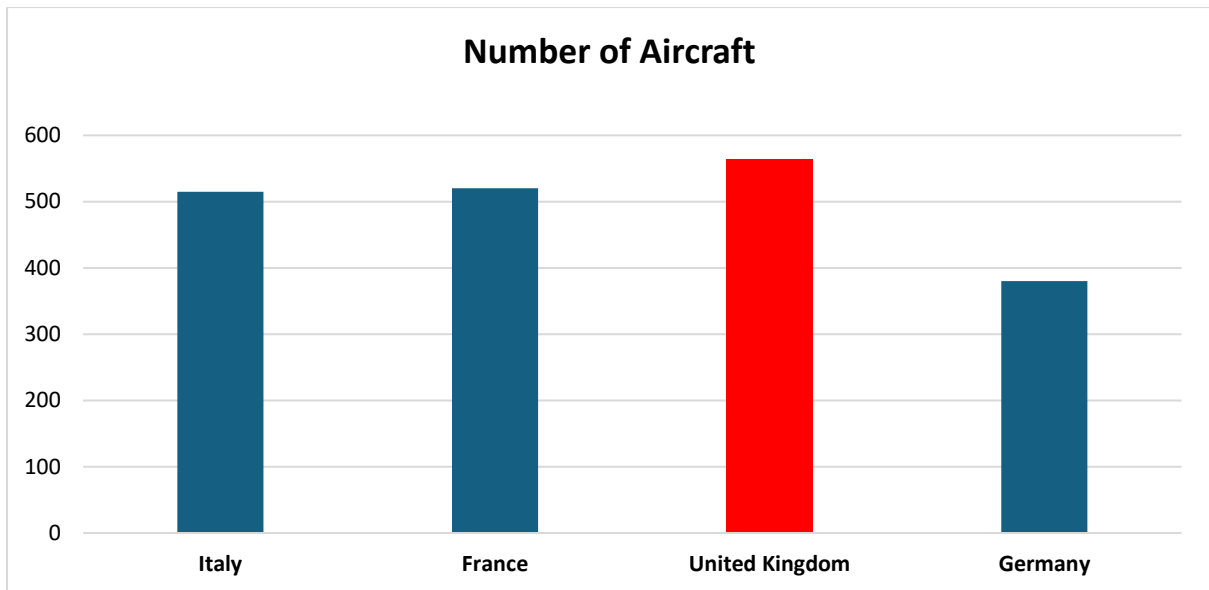


Figure 7- A graph of the total number of aircraft in service in certain countries (WDMMA, 2025)

Clearly, the RAF is suffering far less from a size deficit compared to other nations than previously thought. The same can also be observed by viewing the personnel sizes of the same nations:

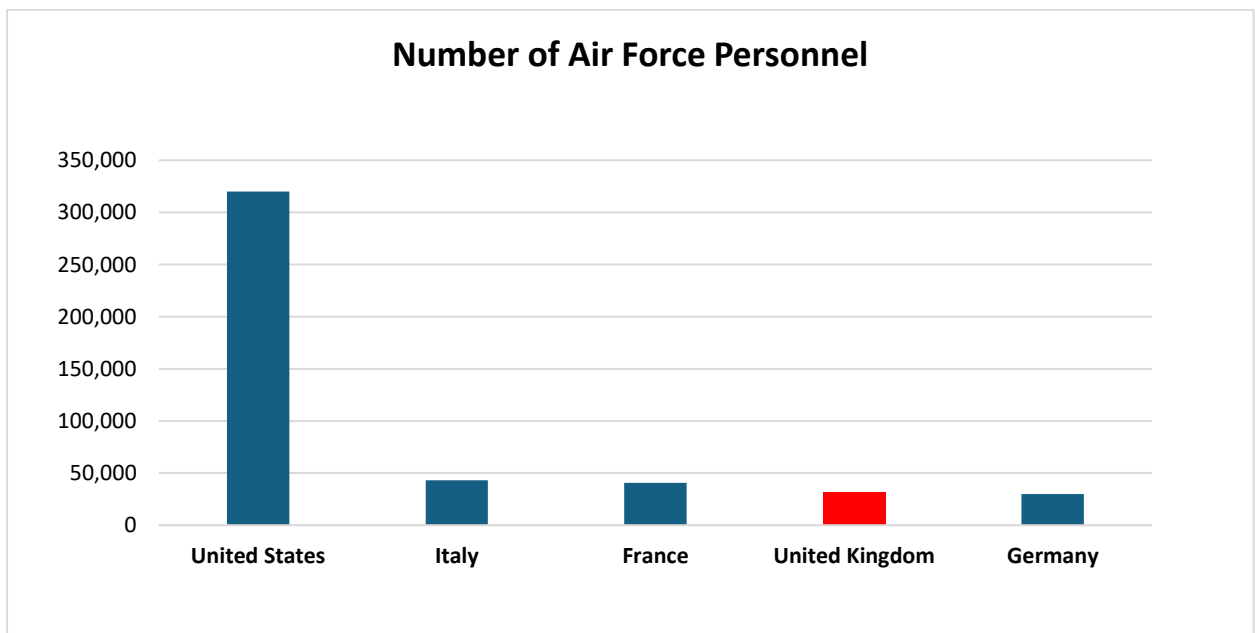


Figure 8- A graph of the total number of air force personnel in certain countries (Seerat Arora (2024))

And without the US:

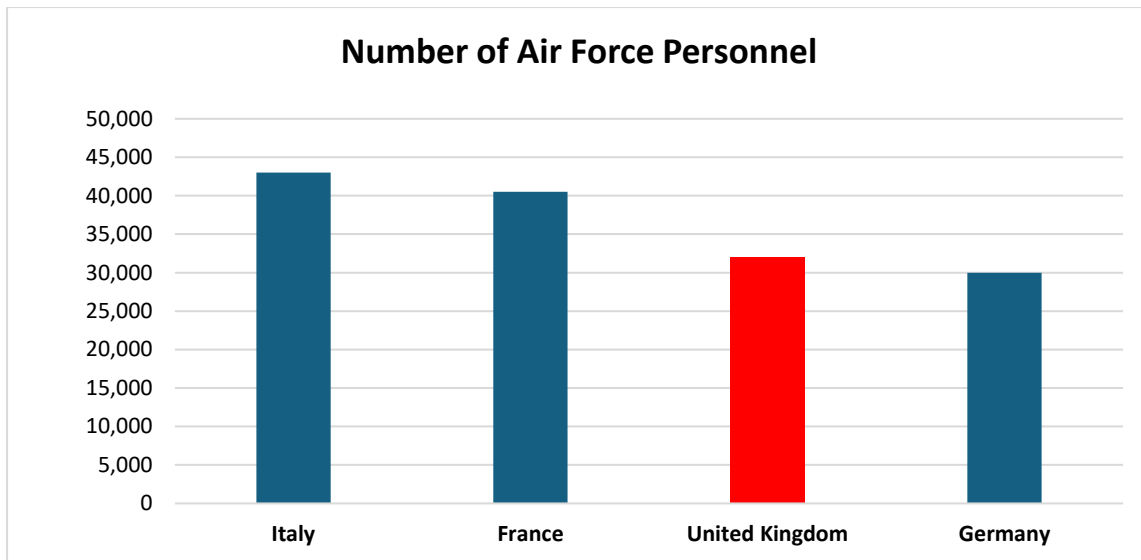


Figure 9- A graph of the total number of air force personnel in certain countries (Seerat Arora (2024))

Even when displaying the proportion of the population employed by a country's air force, the UK is nearly on par with that of the US:

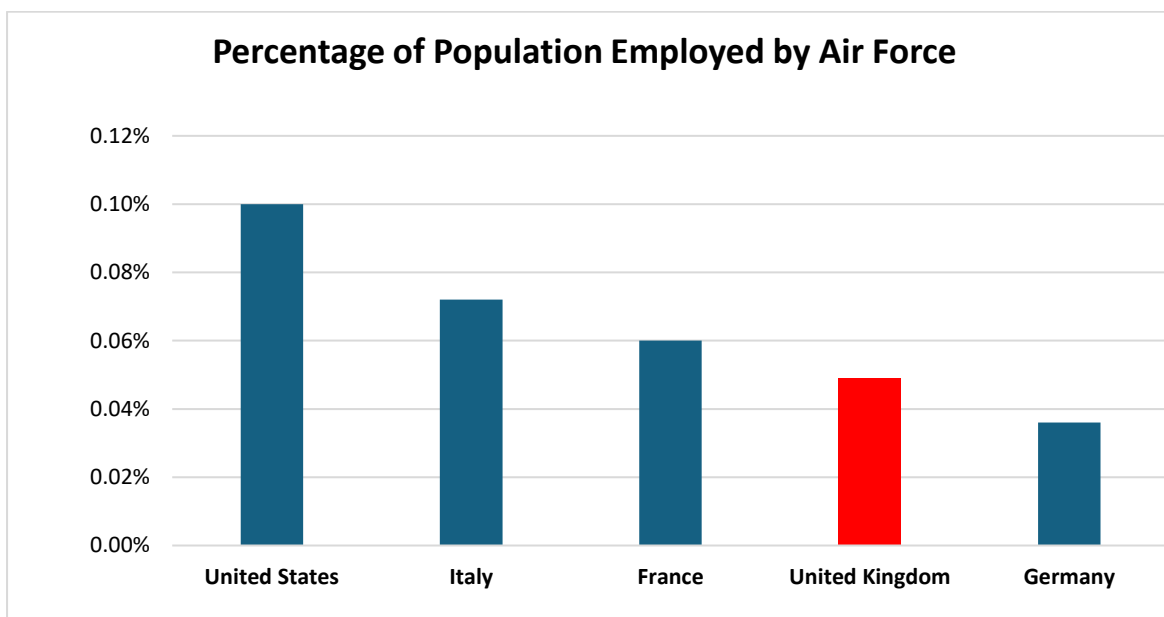


Figure 10- A graph of the percentage of a country's population employed by its air force, in certain countries (Seerat Arora (2024))

Even considering the overall change to air force size since 1990, the RAF is noticeably on par with other allied nations, according to the House of Commons Defence Committee:

	1973	1993	2013	2023
UK	500	450	230	169
France	500	630	290	231
Germany	460	450	209	214
Italy	330	320	220	199

Source: International Institute for Strategic Studies, *The Military Balance*

Figure 11- A table of the number of air force aircraft in service in certain counties (House of Commons Defence Committee (2023))

In fact, using an up-to-date statistic from the House of Commons Defence Committee, an authoritative source on the matter, the UK, compared to most NATO counties, is on the high end of defence spending:

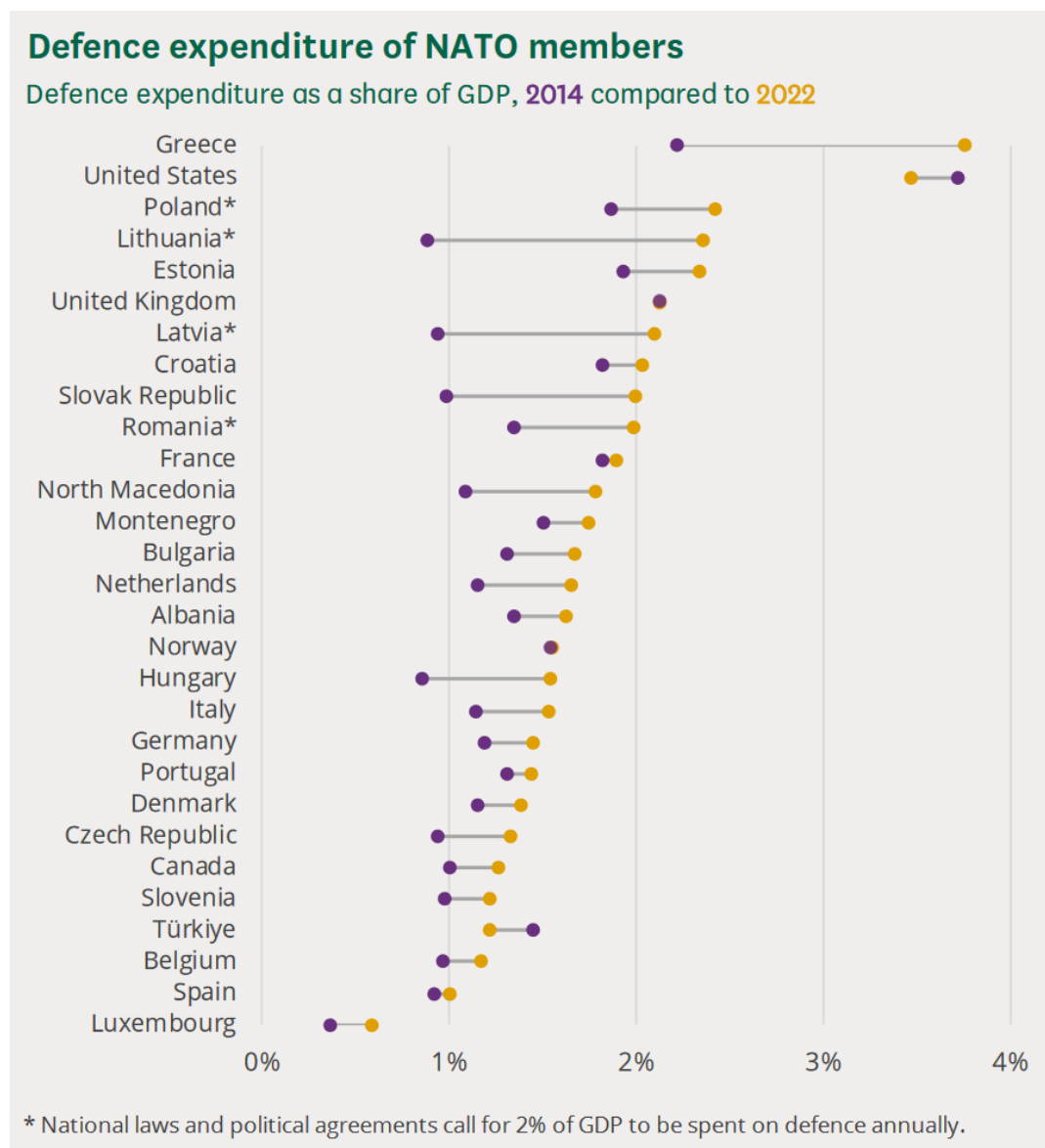


Figure 12- A graph of defence expenditure as a percentage of GDP, in NATO member states (Esme Kirk-Wade (2024))

Clearly, the RAF is on par with many of its allies. Although the RAF is undeniably leaving itself with gaps in capability and shortcomings in overall fleet, size, it seems that many other nations with data openly available are going through similar size reductions, as the world moves away from the possibility of all-out total warfare, negating the need for bloated and oversized air forces.

More importantly, the RAF would almost never fight a conflict on its own. Even if the UK were to find itself implicated in a large-scale conflict against Russia and China, it has historically been able to count on its ally, with a bond forged during the perils of WW2, the United States. Even if the UK and US were to go toe-to-toe with Russia and China, the graphs below suggest that any hypothetical conflict between these nations would likely be quite one-sided:

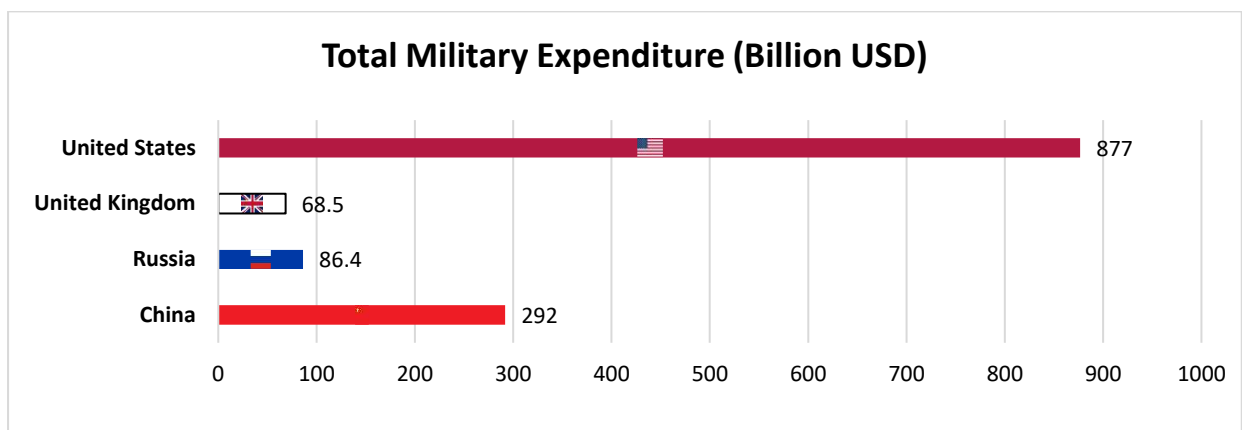


Figure 13- A graph of total military expenditure in certain countries, in billions USD (Stockholm International Peace Research Institute (SIPRI) (2022))

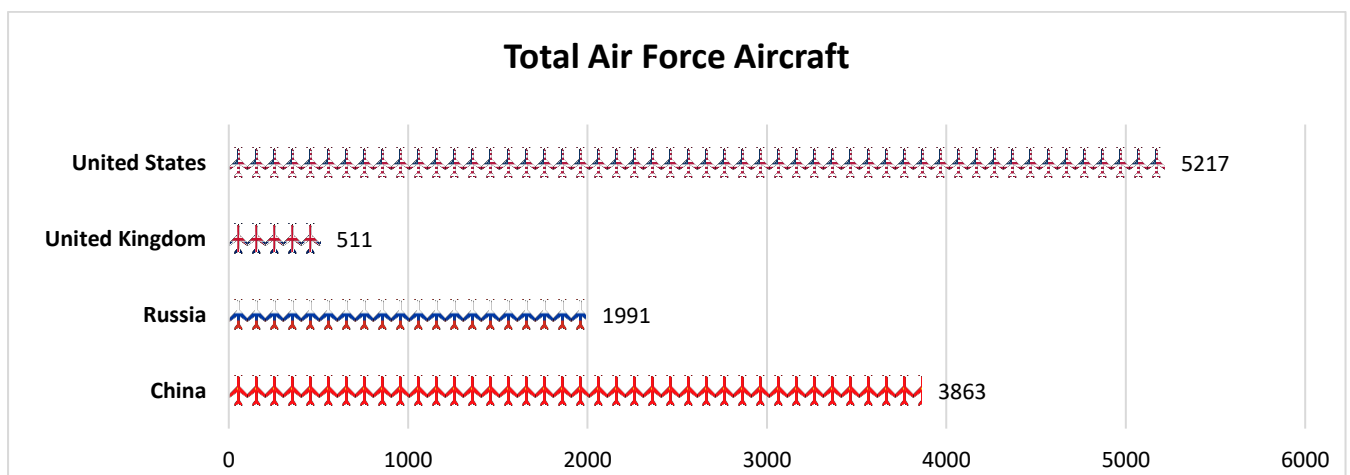


Figure 14- A graph of total number of air force aircraft (WDMMA (2025))

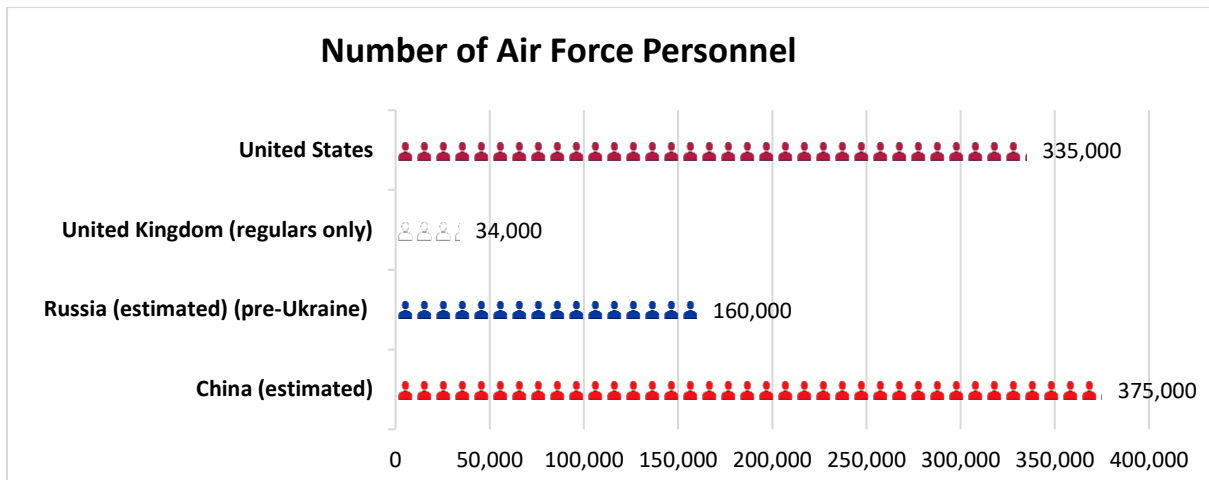


Figure 15- A graph of the total number of air force personnel in certain countries (Seerat Arora (2024))

Furthermore, NATO's Article 5 states, in effect, that an attack against one NATO member state is an attack against all. As such, practically any conflict involving the UK would almost certainly involve NATO too, which holds a majority of military power. Quite literally, NATO is responsible for *a majority* of international defence spending, as can be seen below:

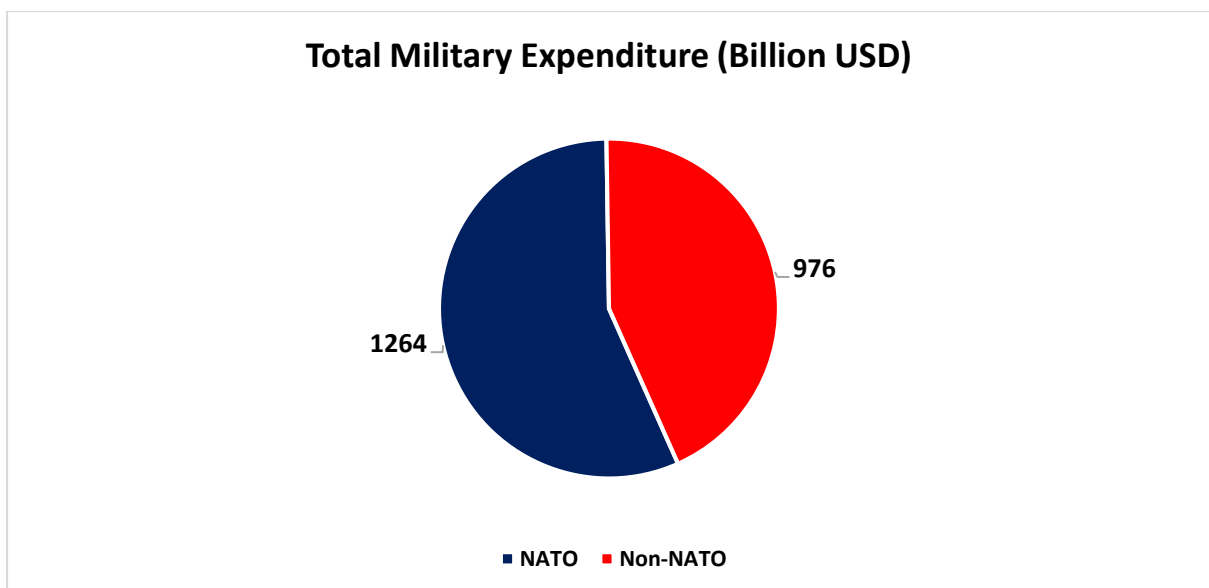


Figure 16- A graph of total military expenditure in NATO vs Non-NATO countries, in billions USD (NATO Public Diplomacy Department (2024))

It should also be said that non-NATO countries also include allies such as New Zealand, Australia, South Korea, and Japan, meaning that any major global conflict would be even more one-sided in favour of the UK. Furthermore, by comparing NATO to China and Russia, we see an even bigger size advantage:

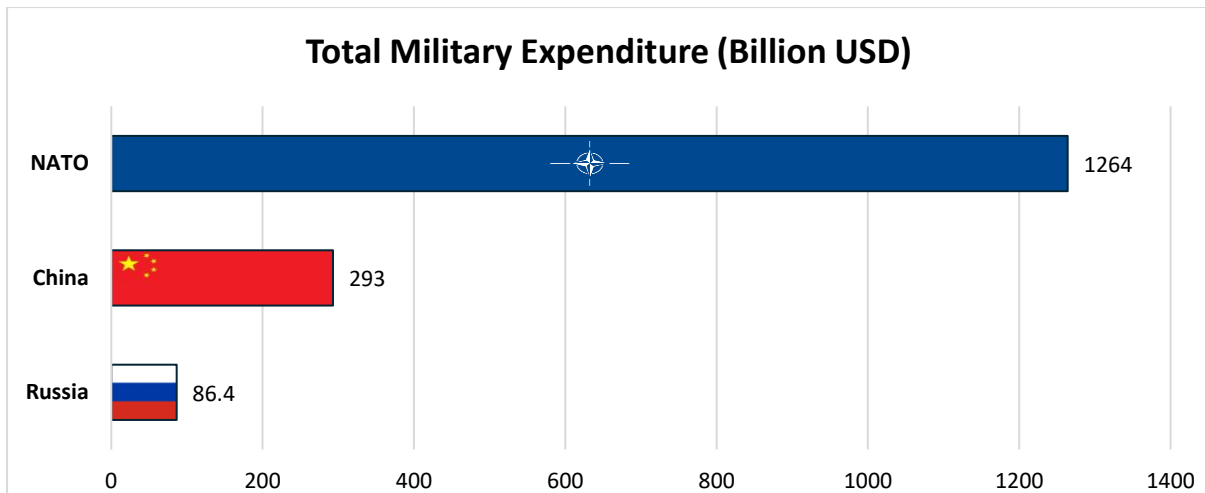


Figure 17- A graph of total military expenditure in NATO countries and potential NATO adversaries, in billions USD (NATO (2024)) (SIPRI (2022))

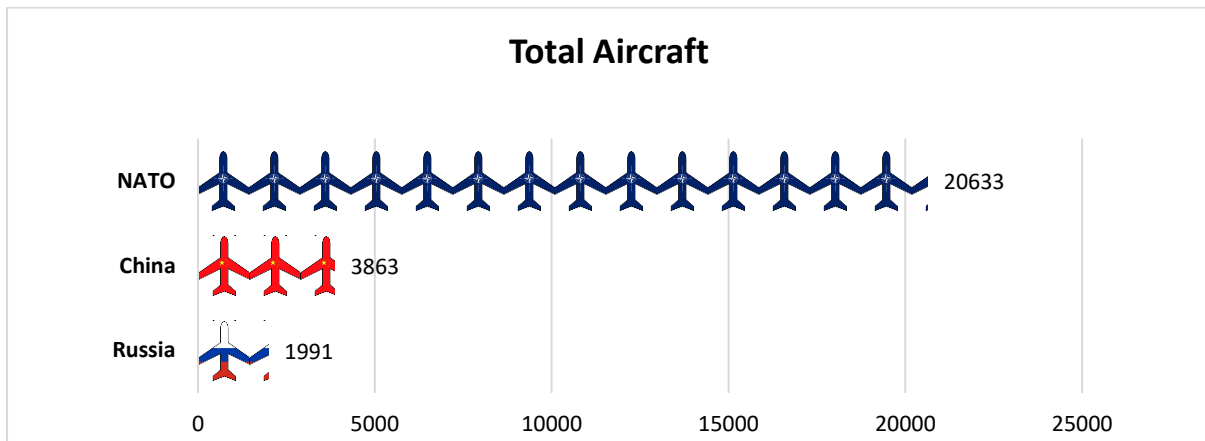


Figure 18- A graph of total aircraft numbers in NATO countries and potential NATO adversaries (WDMMA (2024))

This discrepancy is not only present in terms of quantity, but also in quality. While NATO is in possession of over 1000 battle-tested F-35s, the most advanced and interoperable fighter jet ever produced, Russia and China are in possession of around 230 J-20s and 22 Su-57s between them, which have had limited combat deployment. NATO also possesses 615 tanker aircraft, hugely expanding their reach, and 970 ISR aircraft, enhancing their operational effectiveness, compared to Russia and China's combined total of 40 tanker aircraft and 157 ISR aircraft. More importantly, NATO has been involved in several recent and successful military campaigns, unlike China, who's last operational deployment was in their 1979 defeat to Vietnam, and Russia, who are currently involved in a brutal war of attrition due to their invasion of Ukraine.

Of course, there must be something to say about the recent re-election of Donald Trump in the US, with his repeated calls to end the war in Ukraine *"within 24 hours"*, and his threats to not protect NATO allies if they do not spend the required 2% of GDP on defence. His aims are to focus on peace for Ukraine, rather than enabling it to reclaim all the territory it owned before, according to a former adviser (Sofia Ferreira Santos (2024)). As argued by a defence source in a statement to the BBC, *"it doesn't make sense for Europe's defence interests to be dependent on a few thousand votes in Pennsylvania"* (Laura Kuenssberg (2024)). It is entirely possible that this reliance on the USA is unsustainable, with American public sentiment turning away from a role of global

moderation in favour of an 'America First' approach to geopolitics. These sources, from reliable news websites, use up-to-date information and expert analysis to come to somewhat startling conclusions about the future role of the USA in NATO.

However, it could be that this is the catalyst for Europe increasingly collectivising its military, and ending its reliance on the US, allowing it to more securely protect its own interests without the approval of the US. Although it is highly unlikely that diplomatic relations between Europe and the USA will be irreparably harmed by the incoming president's administration, it is entirely possible that the USA reduces its involvement in European defence affairs, meaning that Europe is forced to increase its internal military collaboration, therefore building closer ties.

Of course, despite all of the threats from President-elect (at time of writing) Trump in his 2017-2021 administration, the US military did not substantially withdraw units from NATO. It could be that his threats from his most recent presidential campaign are simply empty threats of hyperbole to encourage other NATO countries to take a more active role in the organisation. For now, NATO is still as strong as an institution as ever.

Even if the worst-case scenario for NATO were to occur and the UK would be forced to fight a large-scale conflict with minimal help from its allies, the RAF may not be in as pessimistic of a situation as previously thought. Even comparing just the UK to Russia and China, the UK has a far better balance of aircraft types in comparison to the other nations:

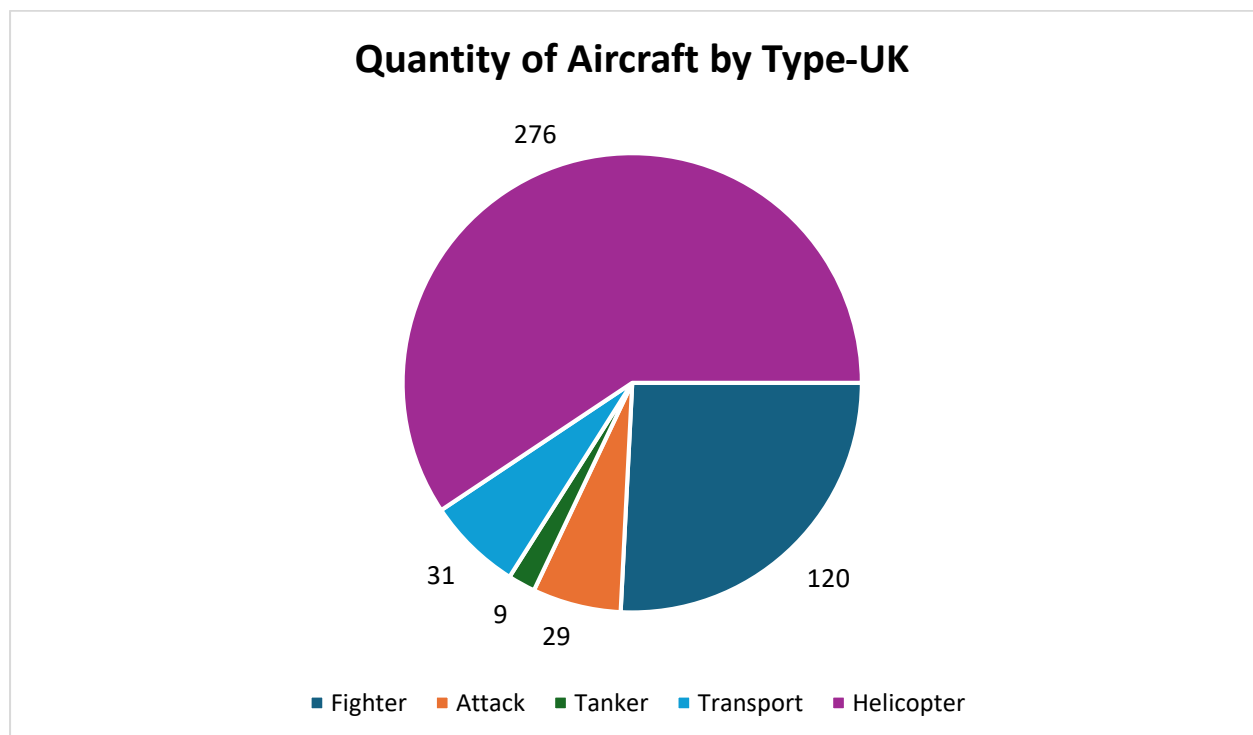


Figure 19- A graph of the composition of the RAF's fleet by aircraft type (WDMMA (2024))

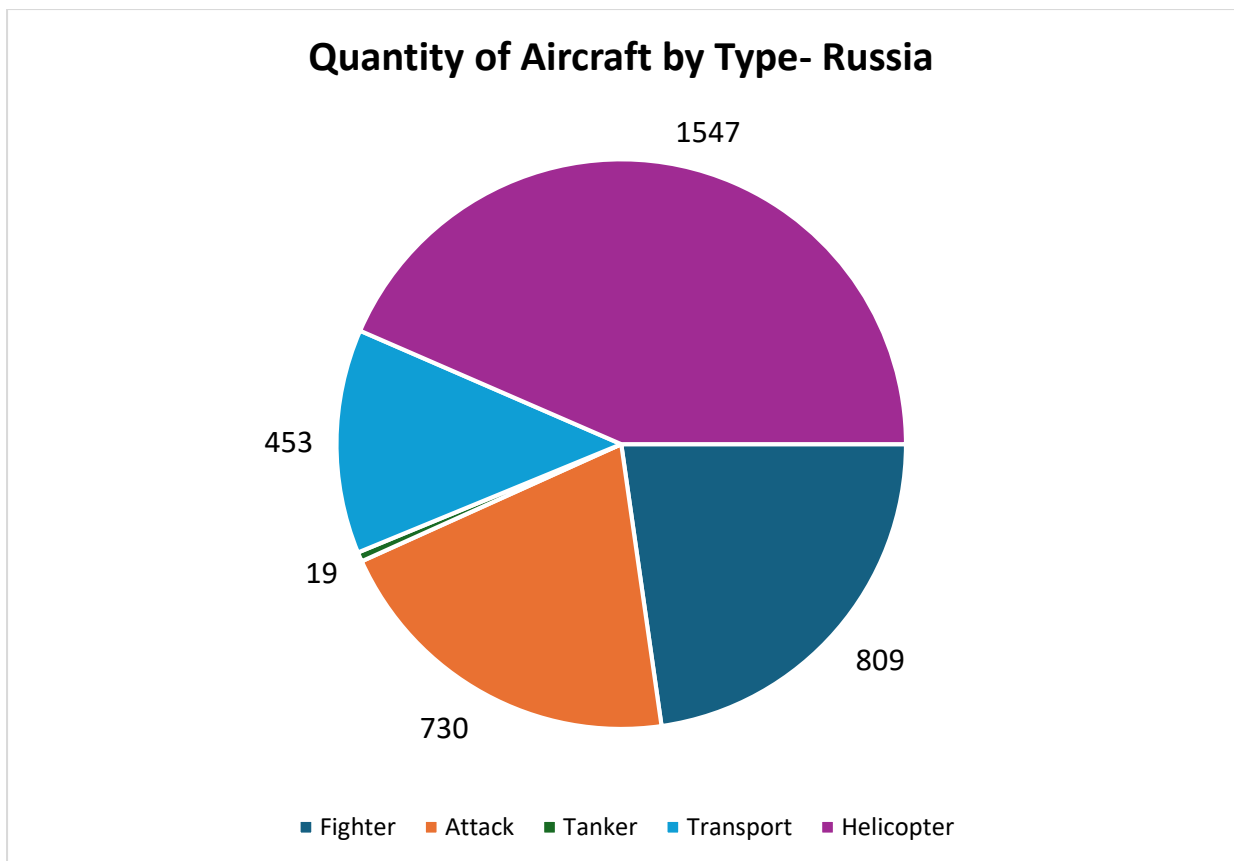


Figure 20- A graph of the composition of the Russian Air Force's fleet by aircraft type (WDMMA (2024))

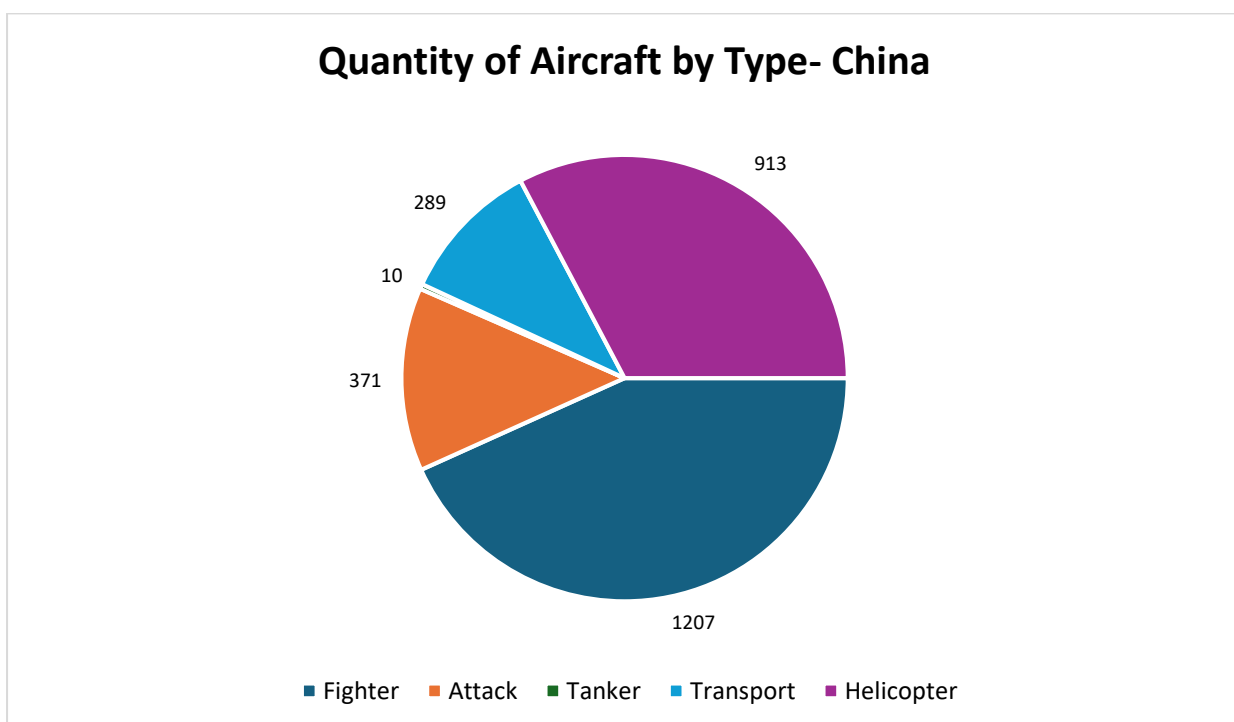


Figure 21- A graph of the composition of the People's Liberation Air Force's fleet by aircraft type (WDMMA (2024))

This fighter-heavy balance that the Russian Air Force and People's Liberation Air Force employ are, very simply, outdated for the sort of scenarios that many air forces are involved in today. Very rarely do bordering first-world nations enter a conflict with one another; Simply put, most conflicts today are expeditionary in nature and require careful logistical movements of troops, equipment, and munitions. As such, this fighter-heavy balance of aircraft in the fleet of these air forces will be a logistical and strategic nightmare to manage. Such a low ratio of transport and tanker aircraft per fighter aircraft will put the Russian and People's Liberation Air Forces at a major strategic disadvantage, as they will face major logistical challenges moving the bulk of their fleet to a distant battlespace. The power of an air force comes from its height, speed, and reach, and with so few tanker and transport aircraft, the reach of these air forces will be hugely reduced, and their respective areas of influence shrunk.

So far, this project has only investigated the effectiveness of the RAF from the perspective of size and strength. Ostensibly, that should be the only way to judge the effectiveness of a military branch- through its ability to project power. However, no organisation will be able to survive long-term without ensuring the personnel who operate within the organisation feel sufficiently respected, fairly treated, and valued.

The RAF has clearly established values it expects itself and its personnel to follow, these values being RISE- Respect, Integrity, Service, Excellence. These values, according to the former Chief of the Air Staff Air Chief Marshal Sir Mike Wigston, *"underpin our ethos and form the foundation of the mutual trust, leadership and teamwork on which our Service depends."* He also stated that he expects that the RAF *"must continue to improve what we offer to you [employed RAF personnel]: continuing to develop your terms of employment to meet your needs, aspirations and expectations; and being truly diverse and inclusive"*. (Royal Air Force (2019)). Clearly, the RAF expects exceptionally high standards from its personnel, and expects to give the same to its personnel in return.

However, recent scandals directly implicating the RAF have suggested that the values of the RAF are not obeyed by all. A number of controversies have resulted in negative press attention directed the RAF's way, including unlawful discrimination, and the abhorrent treatment of female personnel in some cases.

From 2020 to 2021, the RAF unlawfully discriminated against white males in its recruitment practices, in a bid to bolster diversity within the force. The head of recruitment at the time, Group Captain Nicholl, pushed back against the policy, which directly implicated the Chief of the Air Staff at the time, Air Chf Mshl Mike Wigston, who allegedly misled Parliament about the legality of this recruitment practice. This ultimately resulted in Gp Capt Nicholl being forced to resign, instead of implementing a policy believed by higher-ups to be a form of positive action, which she (correctly) identified as positive discrimination, after seeking legal advice on the matter. As a result of these recruitment practices, which breached the Equality Act 2010, 31 men have been identified as missing out on £5000 joining bonuses (Deborah Haynes (2023)). Ultimately, the controversy and following inquiry into the matter created substantial negative sentiment towards the RAF, requiring an apology from the incoming Chief of Air Staff, Air Chf Mshl Sir Richard Knighton (Royal Air Force (2023)).

Even more shockingly, in 2023, an investigation found that the RAF Red Arrows, made up of some of the most exceptional pilots in the British Armed Forces, perhaps even the world, had found that predatory behaviour towards women was *"widespread and normalised."* As reported by the BBC, an investigation found that between 2017 and 2021, numerous cases of unacceptable behaviour occurred, including *"unwanted physical contact, unwanted text messages of a sexual nature, unwanted invitations to engage in sexual activity and "male sexual entitlement" towards women, who were "being viewed as 'property' of either individuals or the squadron"*, which even went as far as the *"exposure of genitals"* on two separate occasions. This ultimately resulted in two serving Red Arrows pilots being dismissed from the force, and a further five team members out of the team of 120 faced *"administrative actions"* (Jonathan Beale, Jemma Crew (2023)). Once again, Air Chf

Mshl Knighton issued a “*unreserved*” apology, and the Red Arrows were forced to fly the next season with only 7 pilots instead of the usual 9, putting on full display to the public the extent to which this toxic environment within the team had affected the capability of the RAF’s most famous display team.

These revelations are, to put it bluntly, shocking. As seen before, the RAF prides itself on being a disciplined organisation, with the values of RISE reflected through its actions. However, these scandals, along with others, such as the MOD housing controversy, have garnered significant amounts of deserved press attention and condemnation.

In order to fully investigate the change in public opinion towards the RAF these scandals had caused, further primary research was required, which was collated through the use of a survey. This survey was sent to people in Years 12 and 13 and encouraged to be shared further by participants.

I obtained a total of 16 responses, 13 of which could be considered ‘appropriate’ for analysis, i.e. the answers provided were not inappropriate or deliberately outlandish. As a result, these results should not be used to draw sweeping generalisations about the opinion of the general public but are a good indication into how people’s opinions might have changed due to these controversies. As such, they should be analysed in conjunction with other research methods, and not analysed independently. This has been achieved by interviewing Mr A Ashment, a music teacher, army reservist, and current lead for the Bishop of Llandaff CiW High School Combined Cadet Force (CCF). Not only can his answers provide expert insight, they should also help to highlight any occurrences where Hawthorne bias (Where a person will respond differently if they are aware they are under observation) has affected the answers of the survey respondents.

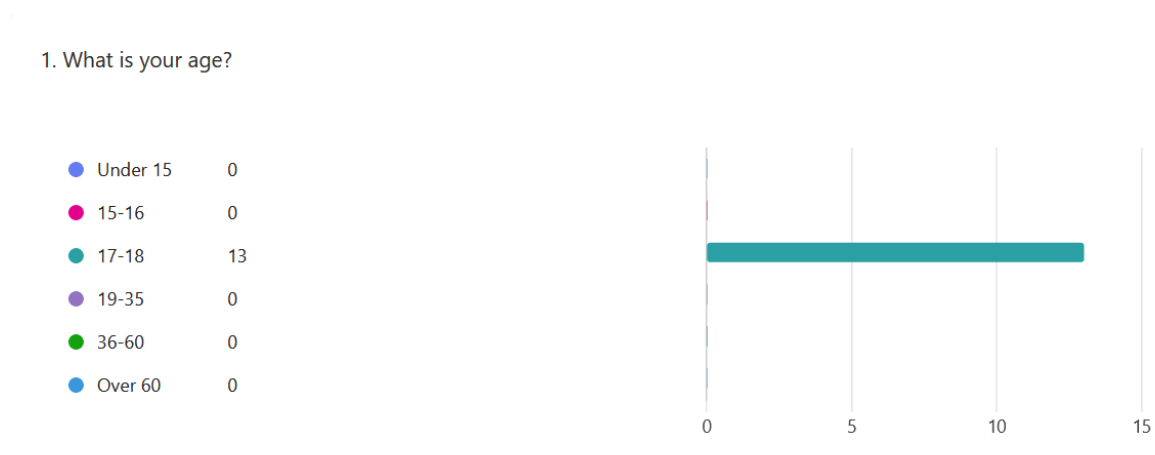


Figure 22- A graph of the age ranges of the respondents of the survey

As can be seen by Figure 22, all of my respondents were 17-18. Although this further affects the reliability of this data, it does mean that I have more accurate data within the ‘young adult’ age range. As such, the obtained results can be used to better understand the mentality of this age group, which could arguably be more beneficial than having a variety of age ranges, given that 17-18 year olds are nearing voting age, and will therefore be able to have a large impact on the capability and future of the RAF, by voting for political parties either supportive or sceptical of the RAF.

2. "The RAF is a net force for good." To what extent would you currently agree with this statement?

[More details](#)



Figure 23- A graph of the average opinion on the RAF, according to survey respondents

3. Why did you give this answer?

[More details](#)

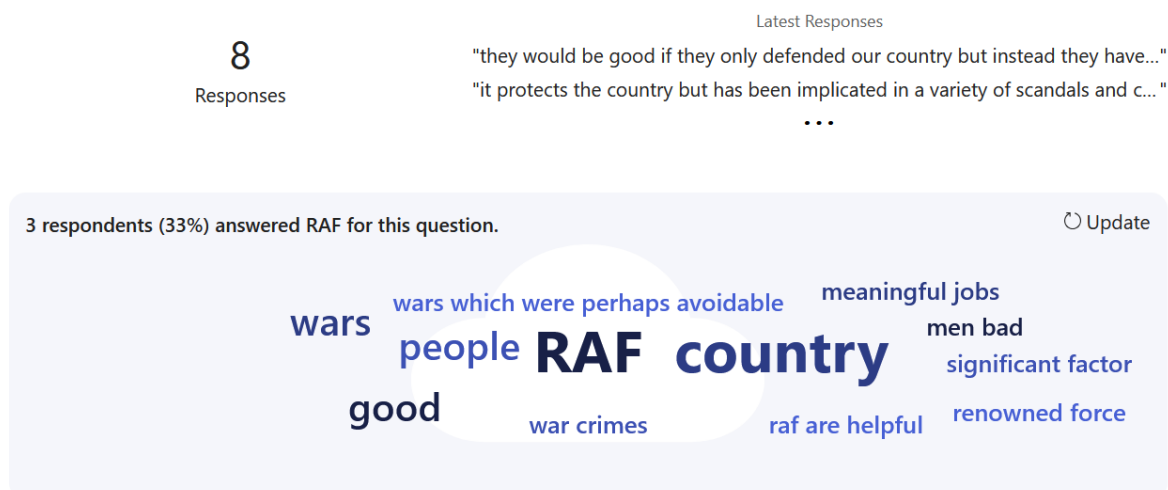


Figure 24- Reasons provided by survey respondents for their opinion on the RAF

Figure 23 shows that the survey respondents have some doubts about the positive influence that the RAF has. Most of these reasons, according to Figure 24, were due to scandals, unnecessary involvement in conflicts, and speculation about illegal actions taken during conflict.

Mr Ashment agreed with the existence of this sentiment. When asked whether these people could have formed their opinion of the RAF from seeing publicised conflicts, such as the wars in Iraq and Afghanistan, through the news growing up, he agreed that this could be a factor in this negative sentiment. He also argued that this negative sentiment could be as a result of narratives being "stretched" by the media, possibly to increase viewership, at the expense of being completely ruthless, such as the Daily Mail article discussed at the end of Aim 1.

He also argued that these results could be as a result of the demographics of the people who completed the survey, particularly their age, meaning that “they [the respondents] are not as knowledgeable about what the RAF has to offer”. Additionally, he stated that he found it “interesting” that the overall sentiment of responses was that “the forces is not good”, stating his disagreement with this sentiment from a “public safety” perspective.

It should also be noted that there was some support given towards the RAF from the respondents, saying that the RAF is “helpful”, provides “meaningful jobs”, as well as providing useful intervention in foreign conflicts. Therefore, it is possible that this variety in opinion is due to personal differences in the need for foreign militaries to intervene in international/expeditionary conflicts.

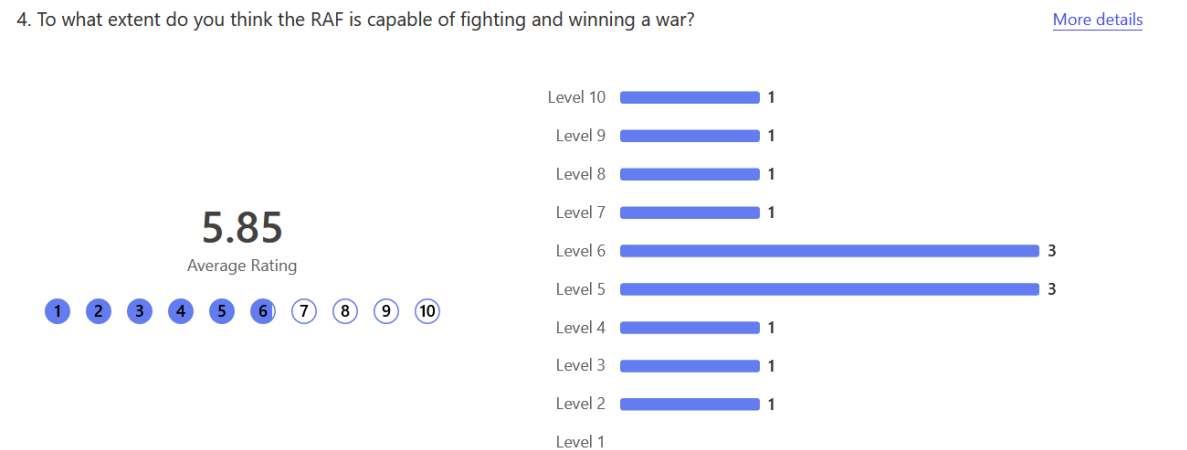


Figure 25- A graph of the average opinion on the RAF’s capability, according to survey respondents

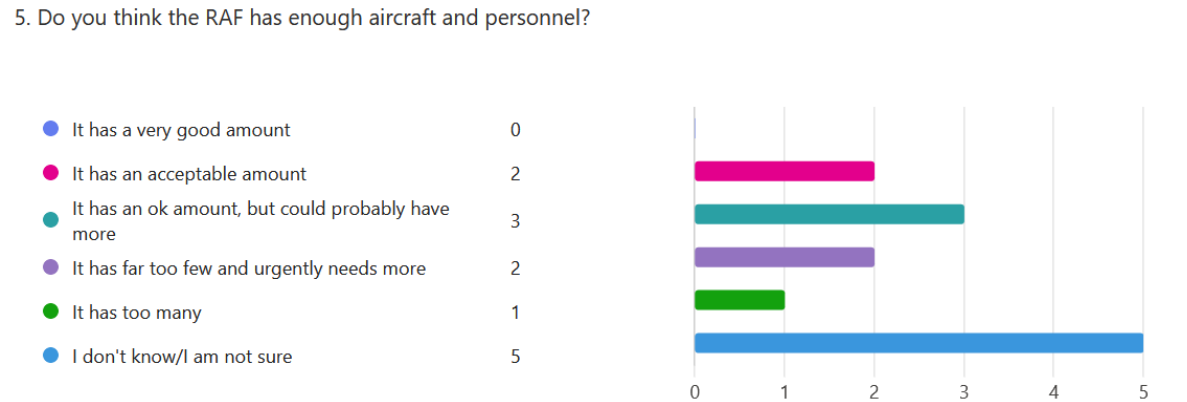


Figure 26- Opinions on the current fleet size of the RAF, according to survey respondents

Figure 26 show a narrow consensus that the RAF has too few aircraft, albeit the category with the majority vote is ‘unsure’, suggesting that my respondents, and possibly this age range in general, are not fully aware of the capability of the RAF. Interestingly, there is a discrepancy between the average rating given in Figure 23 compared to Figure 25, showing a clear difference in opinion between how effectively the RAF operates and

how ethically it operates. This could be as a result of the highly publicised wars in Afghanistan and Iraq mentioned earlier, since, although the deployments as a whole were controversial, there is very little doubt that the RAF achieved their military objectives.

Mr Ashment holds concerns with the RAF’s current capability, saying that “we often hear about military personnel being sent into operational zones now, with limited resources, which is a huge concern”, continuing by explaining that “huge financial strain” was the reason for this. He further stated that this had resulted in cuts to “regiments, fleets, aircrafts, whether its sky, land, sea warfare”. However, he also made the point that the RAF, along with the MOD, is adapting to the restrictive financial situation it finds itself in, saying “Army, RAF, Navy, are still doing the best they can do with limited resources”, with the security of the UK “at the heart of why it’s being done”.

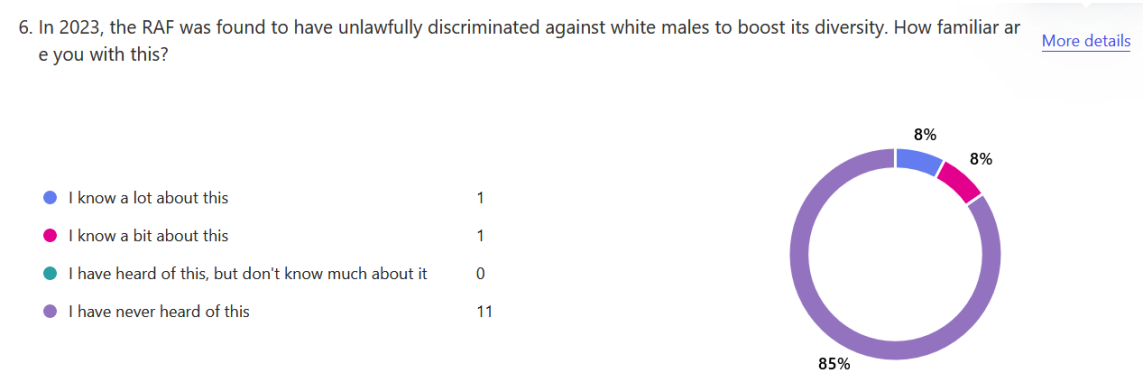


Figure 27- The proportion of respondents who are aware of the 2023 recruitment scandal

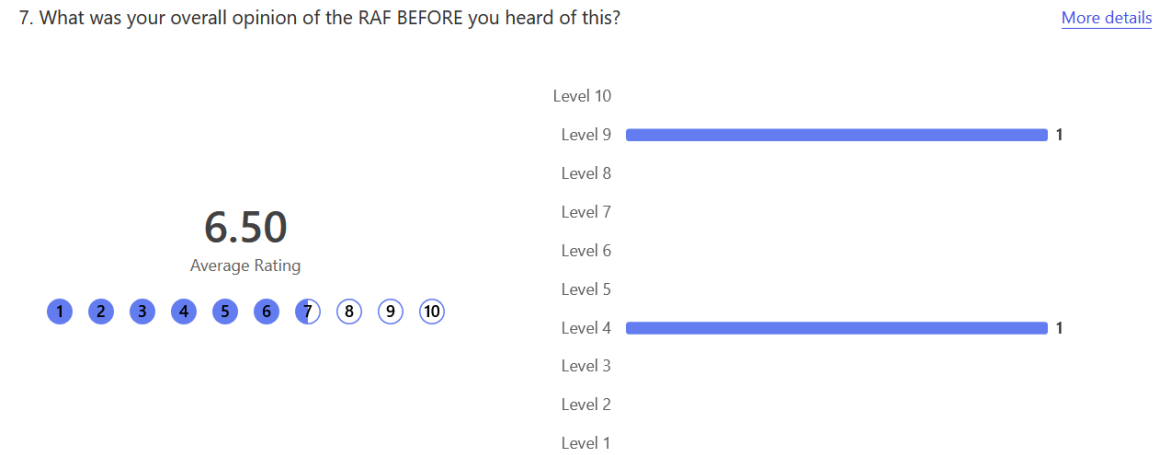


Figure 28- Opinions of survey respondents on the RAF prior to learning about the 2023 recruitment scandal

8. What was your overall opinion of the RAF AFTER you heard of this?

[More details](#)

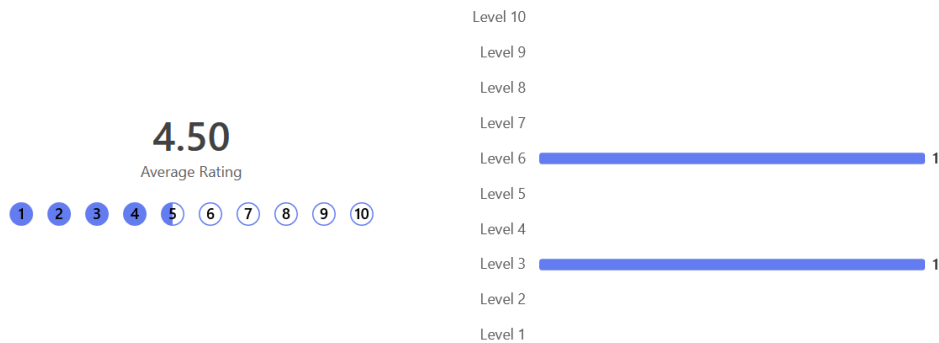


Figure 29- Opinions of survey respondents on the RAF after learning about the 2023 recruitment scandal

Figures 27 shows that among those who responded to the survey, few were aware of the occurrence of the 2023 recruitment scandal. However, given that Figure 28 shows that those who were aware of the scandal had a more positive view of the RAF than the average given in Figure 23, it is likely that these people who answered 'I know a bit' or 'I know a lot' have more interest in the RAF. This scandal is ultimately more concerning for the RAF as a result, as it means that people interested in the RAF are gaining a more negative view of the RAF as a result of it. However, it should be noted that given this response is from a small proportion of an already small pool of people who completed the survey, the exact extent to which public opinion is affected would require further research.

9. In 2023, the Red Arrows came under significant controversy for predatory behaviour being widespread and normalised . How familiar with this are you?

[More details](#)

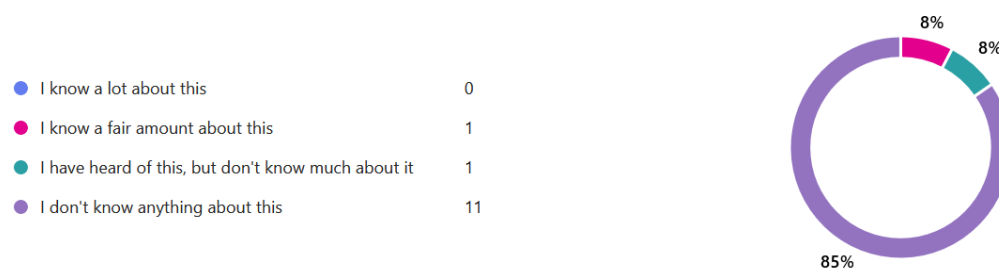


Figure 30- The proportion of respondents who are aware of the 2023 Red Arrows scandal

10. What was your overall opinion of the RAF BEFORE you heard about this?

[More details](#)

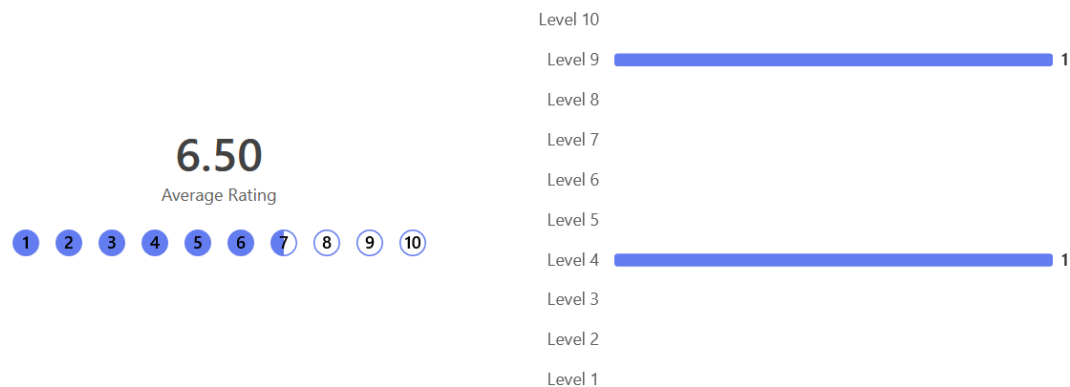


Figure 31- Opinions of survey respondents on the RAF prior to learning about the 2023 Red Arrows scandal

11. What was your overall opinion of the RAF AFTER you heard about this?

[More details](#)

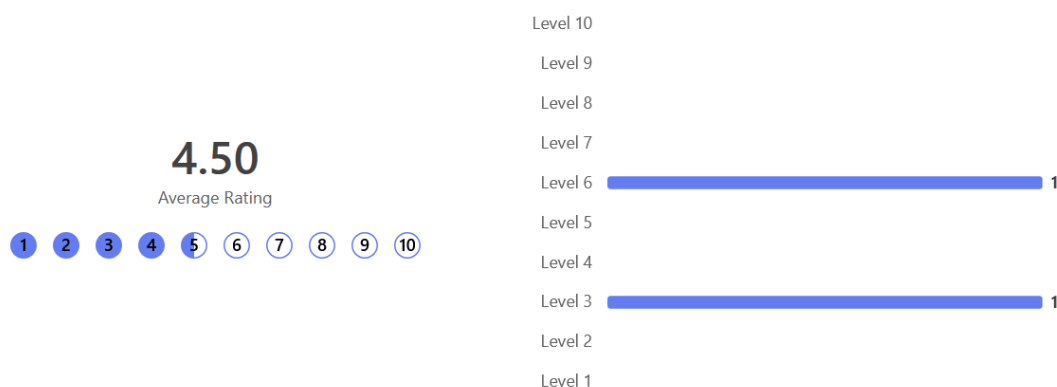


Figure 32- Opinions of survey respondents on the RAF after learning about the 2023 Red Arrows scandal

Similarly to Figures 27 through 29, Figure 30 shows an overall lack of awareness around the 2023 Red Arrows Scandal. However, this group of questions was answered almost identically to the previous group of questions, again suggesting that the respondents who may hold some interest in the RAF are seeing the force in a more negative light.

Once again, it should be noted that given this response is from a small proportion of an already small pool of people who completed the survey, the exact extent to which public opinion is affected would require further research.

When questioned about the scandal, Mr Ashment highlighted disagreement towards the notion that the UK armed forces and the MOD have prevalent occurrences of predatory and otherwise unacceptable behaviours, saying that “the military has done a lot to get on board with diversity, equality, and making sure that people are being treated fairly”.

12. Since 2016, it has been found that the RAF's overall size has dropped substantially, from 724 aircraft to 564. How familiar are you with this? [More details](#)



Figure 33- The proportion of respondents who are aware of the decline in RAF fleet size since 2016

13. What was your overall opinion of the RAF BEFORE you heard about this? [More details](#)

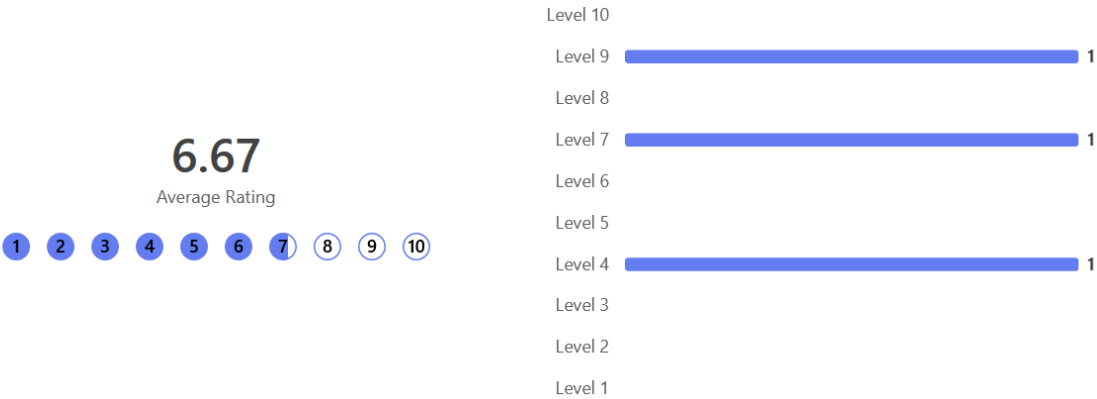


Figure 34- Opinions of survey respondents on the RAF prior to learning about the decline in fleet size since 2016

14. What was your overall opinion of the RAF AFTER you heard about this? [More details](#)



Figure 35- Opinions of survey respondents on the RAF after learning about the decline in fleet size since 2016

Figures 33 shows that the decline in fleet size is the most well-known major controversy the RAF has faced recently, likely as a result of its extensive and consistent press coverage. Once again, the same two respondents who answered ‘I know a bit’ or ‘I know a fair amount’ for the previous group of questions also responded in an identical manner, with the third respondent to answer that they were aware of this story to also note a drop in opinion about the RAF after hearing of the story. Once again, the RAF’s actions are leading to those with some knowledge about the force to change their opinion to be more negative after hearing of the scandal.

Once again, the exact extent to which this scandal has affected public psyche will require more research to conduct, due to the lack of responses to the survey.

15. Has your opinion of the RAF changed after hearing about all of the controversies above?



Figure 36- The proportion of respondents whose opinion on the RAF changed after hearing of the above controversies

16. What is your opinion of the RAF after hearing about all of the controversies above?

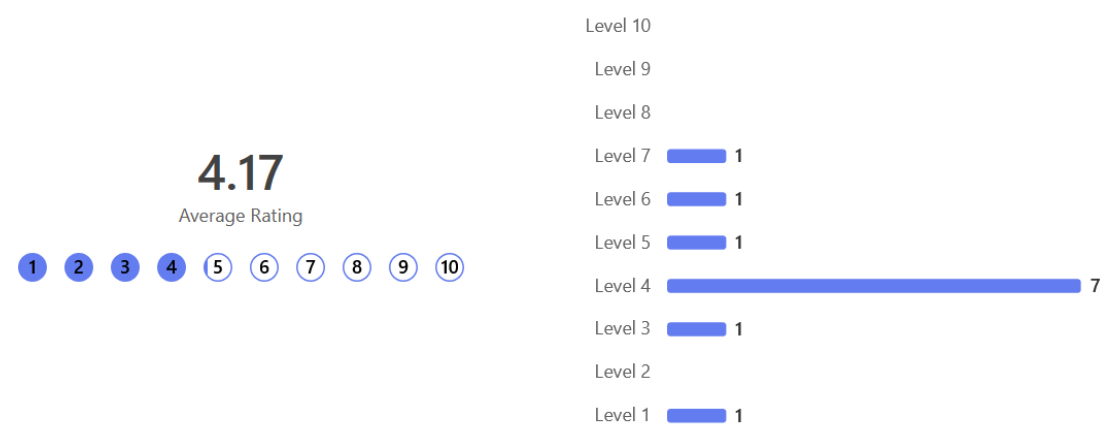


Figure 37- The opinion of respondents on the RAF after hearing of the above controversies

Figures 37 shows that the average public opinion towards the RAF had dropped, in comparison to Figure 23.

To fully determine whether a drop in opinion could be attributed to knowledge of these scandals, a normal hypothesis test was used.

For this test, the null hypothesis (H_0) would be that the mean opinion of the RAF, μ , would be equal to 5.54, or the original average opinion as given in Figure 23. The alternative hypothesis (H_1) would be that the average mean, μ , would be less than 5.54, showing a decrease in opinion which could be attributed to learning about these controversies.

To find the standard deviation, σ , I used an online tool, with the formula for its calculation displayed below. This obtained a value for standard deviation of $\sigma=1.8652854850742$.

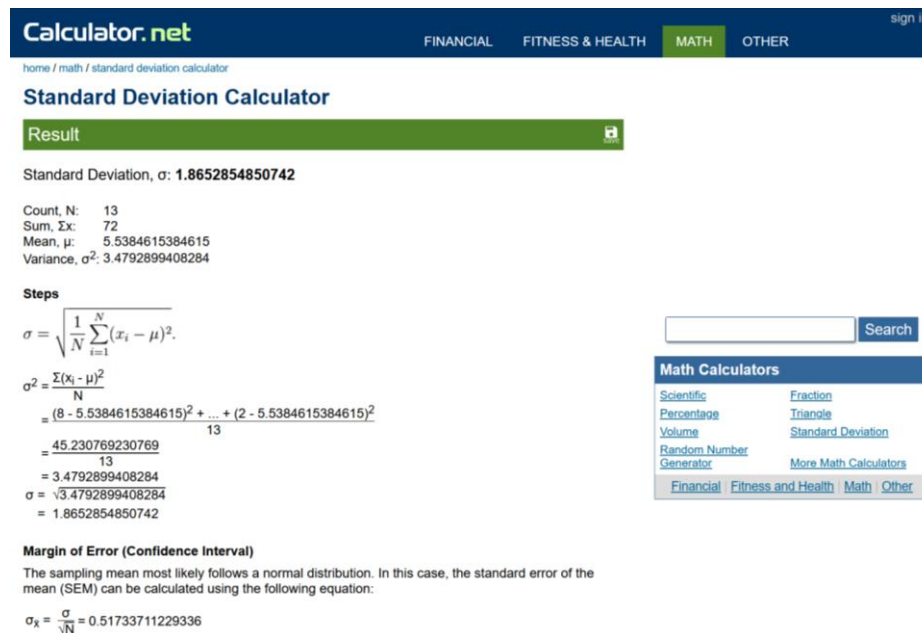


Figure 38- The calculation of standard deviation from the values given in Figure 23

From this point, I used the mean value given in Figure 37 as the sample mean, $\bar{x}$, and described the distribution using $X \sim N(5.54, \frac{1.865^2}{13})$. Using the sample mean, $\bar{X} \sim N(5.54, \frac{1.865^2}{13})$, I found that the probability of the sample mean being less than or equal to the sample mean of 4.17, $P(\bar{X} < 4.17) = 0.00404$, therefore giving sufficient evidence to reject the null hypothesis. In effect, this means that there is a 99.96% probability that the mean opinion has shifted. Given that the only factor that could change this opinion was learning about the controversies outlined in the survey, it is beyond reasonable doubt that these controversies have affected the opinions of the respondents.

On the whole, it appears that these controversies have had a substantial impact on the respondents' views on the RAF. Although they may believe that the RAF is still capable, it is clear that not only are young people unaware of the controversies surrounding the RAF recently, but also that these controversies can hugely affect their opinion of the force.

Of course, based solely on Aim 1, it would be easy to assume that the RAF is operationally ineffective. The majority of news stories on the matter of the RAF's, and indeed the entire British military's, strength have not been very positive in nature. It seems that the majority of the press coverage the RAF receives mentions only the gap in airborne capability (Deborah Haynes, 2023) (BBC, 2019) (Richard Thomas, 2022). Furthermore, most

estimations of the power of the RAF in comparison to other armed forces place it below the average power of the entire British Armed Forces (Sinead Baker, Thibault Spirlet, 2023) (Sinead Baker, 2023). However, as mentioned before, sheer numbers are no guarantee of success. The RAF is aiming to build an extremely modernised and interconnected fleet, even if that means sacrificing quantity for quality. Additionally, as threats more increasingly digital and our most valuable assets are now digital, instead of physical, there has been a greater push to protect these assets from malicious state and non-state actors.

In order to fully recognise how the RAF is shifting to becoming a more technologically advanced fighting force, we must discuss its tooth-to-tail ratio. Put simply, this is the ratio of combat personnel and equipment (i.e. Pilots, RAF Regiment, Fast-Jets) to support personnel and equipment (i.e. engineers, air operations controllers, transport aircraft). Because more technologically advanced equipment requires more servicing and maintenance, due to its increased complexity, a higher tooth-to-tail ratio is better, as it represents a more advanced, and therefore likely more capable, force. Current estimates give the RAF a tooth-to-tail ratio of 1:3, which implies a high level of technological capability. This is partly due to escalating costs of each new generation of aircraft, which has been previously discussed. Additionally, the RAF is placing further emphasis on control of the cyberspace and space domains, aiming to deny, disrupt, and degrade hostile actors. This also allows for better command and control of assets, as described in JDP 0-30, *"Air platforms and systems have become increasingly dependent on cyberspace and the people, processes, technology and data within it."* Its impact in space is also of increasing importance, helping to monitor everything from space debris to ballistic missiles.

Over the past number of years, the RAF is increasingly switching from a single-domain force to a three-domain. Although this could be perceived to have diminished the RAF's control of the air, there is no denying that its influence in cyberspace and space, one far harder to gauge by counting aircraft or personnel, is growing every day.

Aim 3- To Determine the Current Overall Effectiveness of the RAF and How This May Change in Future

As has been previously discussed, the theoretical strength of the RAF is certainly up for debate. Although its fleet size is small, it has a collection of some of the most advanced aircraft created. Although it has nearly the lowest number of personnel in its history, it is deploying them in increasingly effective ways, especially via the use of the 'whole force' concept. Although it is missing some key aircraft types, it has the support of NATO to fill these gaps. Realistically, the only way to determine the effectiveness of the RAF is to look at its historical deployments.

The RAF has been involved in many major deployments over its history, with the most recent deployments being during the Falklands War, and Operations Granby (Gulf War), Veratis, Herrick and Toral (Afghanistan), Telic (Iraq), Ellamy (Libya), Shader (Counter-ISIS), Pitting (Evacuation of Kabul), and Interflex (Support of Ukraine). Additionally, the force has been involved in a number of Operations as part of their role in NATO, such as Operations Azotize (Baltic Air Policing) and Biloxi (Romanian Air Policing), as well as NTM-I, the development of Iraqi security forces. Furthermore, the Force has been involved in a variety of NATO training exercises, such as Exercise Air Defender, Pitch Black, and Nellis Red Flag. Furthermore, it is solely responsible for Quick Reaction Alert (QRA) protection in both the UK and the Falkland Islands, where interceptor aircraft are launched upon detection of illegal airspace entry.

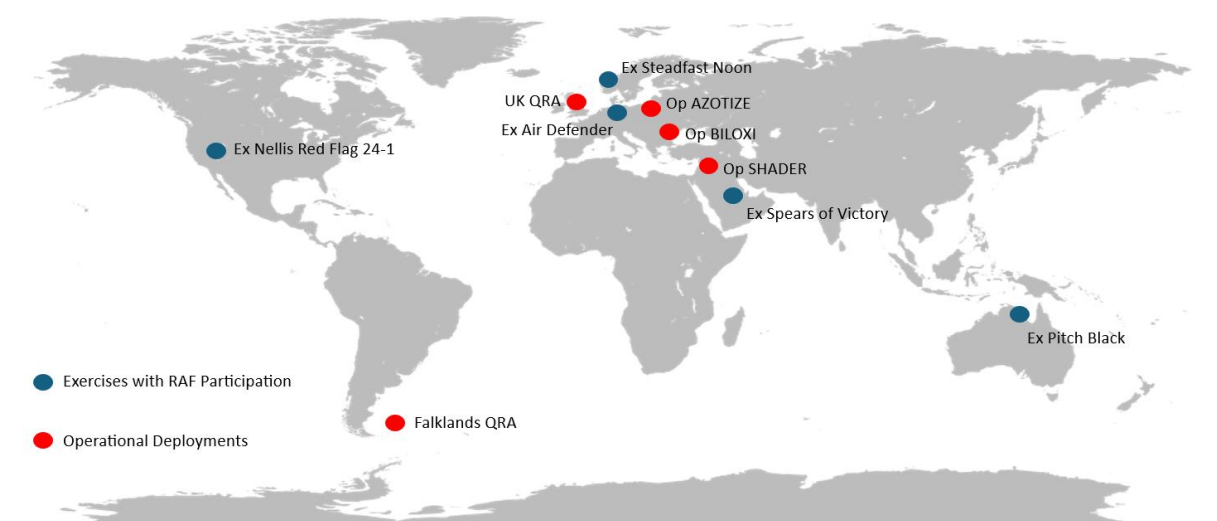


Figure 39- A map of all recent NATO deployments

These deployments help to highlight the operational capability of the RAF in detail. For example, Operation Pitting, the 2021 Evacuation of Kabul, managed to evacuate 5,000 British citizens and 8,000 Afghans from potential persecution over the course of 18 days, in an extremely dangerous and ever-changing environment. Although the RAF was able to rapidly deploy to the area, it still relied on US-led perimeter security and the use of Joint Tactical Air Controllers (JTACs) to support the mission. Despite the fact that it showed the ability of the RAF to some degree through its rapid deployment, the evacuation was also highly controversial, due to its unexpected and somewhat chaotic nature.

Additionally, the RAF is often involved in air policing missions, to protect NATO airspace from foreign threats. Last year, the RAF conducted a 6-month long NATO Air Policing deployment in the Baltics from March to August, which involved 50 separate occurrences of RAF Typhoon aircraft being scrambled to intercept Russian

aircraft, including Fighter-Bombers and ISR aircraft. Although this shows that the RAF can protect its allies, it only has the capacity to deploy up to 4 aircraft on these missions, which may not be enough to provide sufficient protection, if some are undergoing maintenance whilst others have already been scrambled.

Although these operations do cast some doubt on the strength of the RAF, it still emphasises that the RAF holds a strong degree of air power, managing to conduct a rapid evacuation and protect NATO airspace. This indicates that the RAF not only holds a degree of air superiority over Europe and beyond but is also highly interoperable with NATO.

Clearly, the RAF's current abilities lack in comparison to itself even 10 years ago. However, the RAF still has plenty of political support, along with the entire MOD, with the recent announcement by the former Prime Minister, Rishi Sunak, that defence spending is to reach 2.5% of total GDP by 2029, as stated in an up-to-date, politically neutral source from the Government (HM Government (2024)). Additionally, the RAF's current efforts to modernise and reconstruct itself are not without merit.

The new Protector drone, designed to replace the Reaper, is already in service, providing new ISR capabilities, and having the ability to carry modern Paveway and Brimstone missiles.

The Atlas C-1 fleet, despite all its shortcomings in maintenance and abilities, is now in the process of being certified on the niche special forces capabilities once undertaken by the C-130J, removing the issue of lost capability that occurred with the loss of the Hercules (Deborah Haynes (2023)).

The E-7A Wedgetail's expected arrival into service is approaching ever closer, with the aircraft already receiving its paint scheme in preparation. Although the order was reduced from 5 airframes to 3 (House of Commons Defence Committee (2023)), it is still an enormous success that the RAF will have AWACS capability back in its fleet imminently.

Finally, the RAF is continually developing its so-called '6th Generation' aircraft, known as Tempest, planned to replace the Eurofighter Typhoon in 2035. However, the extent to which this timeframe is likely, especially considering the controversies around delays in other aircraft and military equipment, such as Ajax, can be debated.

Most analysts agree that the best way to improve the RAF would be to increase its funding. However, as the RAF is a government agency under the umbrella of the MOD, it cannot directly control how much funding it receives. Importantly, it does have the ability to control how that money is spent, which could lead it to choose to be even more dominant in the space domain, or instead to rebuild its fleet of aircraft. However, it is probably most essential that the RAF removes itself from any sort of controversy, in order to rebuild public trust in itself, therefore increasing support, and the possibility that more of the public budget be allocated towards it.

In order to fully understand whether the public agreed with this sentiment, further primary research was required. Using the same survey from Aim 2, as well as information from the interview with Mr Ashment, it was possible to find the most optimal methods for the RAF to successfully revive its public image.

17. How long do you think it will take the RAF to revive its public image?

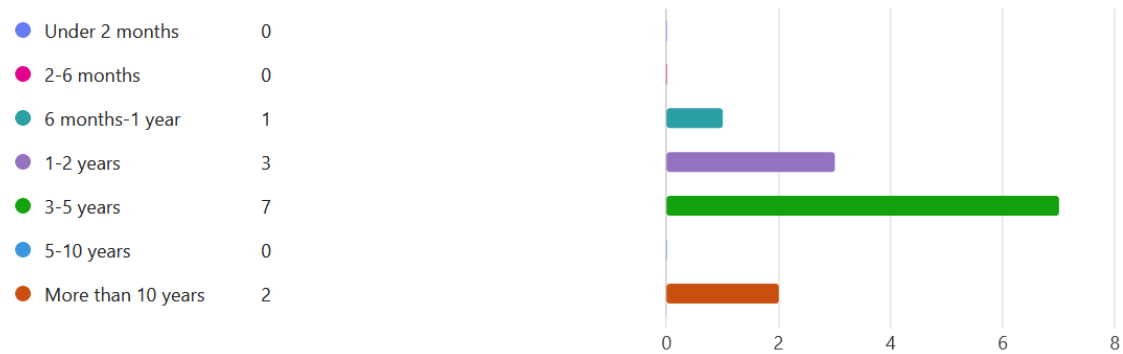


Figure 40- The opinions of respondents on the amount of time it will take the RAF to revive its public image

18. How effective would you think the following methods would be at improving the RAF's public image?

[More details](#)

- This method will not work at all ● This method might work a bit ● This method would be reasonably effective
● This method would be exceptionally effective ● Not sure

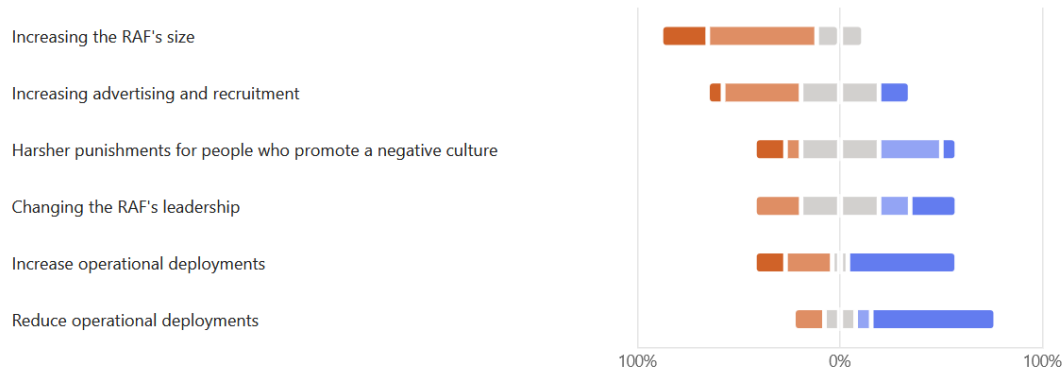


Figure 41- The opinions of the respondents on some solutions to repair the RAF's image

19. Are there any other strategies you would consider effective, and why?

[More details](#)

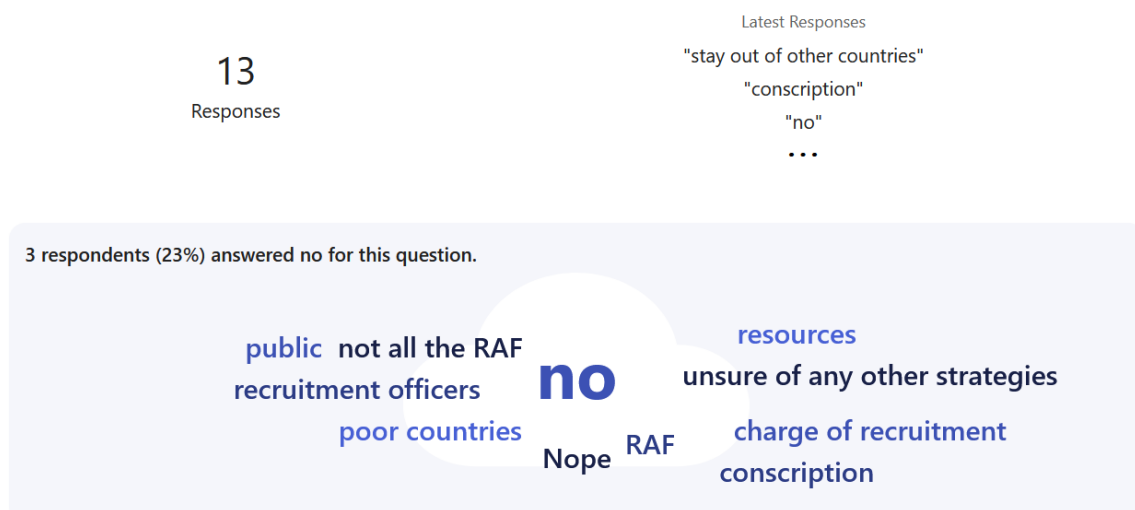


Figure 42- Other strategies to repair the RAF's image provided by respondents

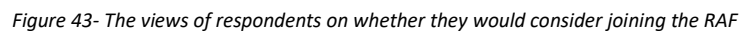
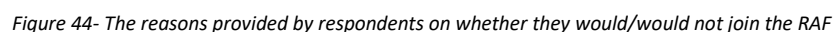
Figures 40 through 42 suggest that the revival of the RAF's public image will be a long-term process, according to respondents, with the modal time range being 3-5 years. Mr Ashment agreed with this sentiment, stating that *"it would certainly be years"*, and that he hoped or the necessary timespan to be within *"single figures of years"*. Although this shows the confidence Mr Ashment has in the ability of the RAF to revive its image, it should also be noted that he has hinted at the possibility the RAF will require 10+ years to outwardly reform itself, in a near-worst-case scenario.

Curiously, the two most popular method of RAF reform, according to Figure 41, are to *'reduce operational deployments'*, and *'increase operational deployments'*. This hints at a very interesting conflict in opinions within the surveyed group, possibly hinting that the idea of military intervention is controversial for the youth of today, which is certainly an issue requiring research. The next most popular suggested methods of reform were based around a change of leadership and harsher punishments for those causing negative cultures. This seems to imply that the respondents were more united on the issue of the culture within the RAF, possibly suggesting that the environment within the RAF produces a toxic and negative environment, according to their opinions. Finally, the least popular methods were to increase recruitment and advertising, and to increase the size of the RAF. These methods were likely seen by respondents as a way to brush these scandals under the rug, as they did not overtly offer a method to eliminate toxicity from the force.

Interestingly, Mr Ashment disagreed with the most effective method, stating that the best way to mellow public opinion was to *"start with kind of schooling and educating primary, secondary, college, university students, or people of that age"*, which suggests that this drop in public support isn't solely due to the recent scandals, but also a lack of understanding around what the RAF does. He emphasised this point by questioning *"maybe the RAF could advertise the fact that they are doing as much as what they're doing- Not just the RAF, but the MOD, they're doing what they're doing, again to protect public security safety"*. He noted how increased activity in the cyberspace and space domains (which is where the RAF is increasingly operating) could have resulted in the lack of awareness, stating *"when we're talking about cyber and security threats, there is an element of security for themselves and [...] it's a breach of security, if we publicise things too much"*.

Mr Ashment also highlighted the roles organisations like the CCF could play in helping to increase knowledge about the RAF, saying that *"us having a unit in school, there is no better way to spread that message of why we're doing what we're doing"*, and continuing by describing the benefits of the cadet services due to *"skills,*

20. Would you ever join the RAF?

[More details](#)

40

Curiously, Mr Ashment stated that there is even more support for those in the armed forces now compared to when he joined, due to the existence of graduate schemes, where *“[the] MOD will sponsor certain students, they will pay for university courses, fees, and then the transition into forces for a certain amount of time”*. This could rectify the reasoning provided by the respondents for not wanting to join the RAF, with many of the reasons given in Figure 44 involving a desire to go to university and concerns about a lack of career advancement. It seems quite likely that were the respondents made aware of schemes like the Defence STEM Undergraduate Sponsorship (DSUS), or the Medical Scholarship, then many of these concerns highlighted by respondents would no longer be an issue.

Clearly, there is some disagreement about the most effective way to revive public image towards the RAF. However, it does seem that the RAF should further develop its ability to share its active taskings to the general public, in order to maintain levels of public curiosity, and potentially interest and support as a result. It could also likely benefit from further emphasising the opportunity for career development and personal progression that can be achieved with employment in the force, as well as further investment into the Cadet Forces.

Finally, the RAF must ensure it can comprehensively and definitively eliminate any sort of culture that could lead to the force being put into disrepute. Clearly, these scandals can hugely negatively influence peoples' opinion on the RAF. As a result, the force must ensure that these sorts of scandals are relegated to the RAF of old, and the current RAF becomes a beacon for opportunity, and a symbol for equality, justice, and security across both the country and the world.

Conclusion

Aim 1

Clearly, the RAF has faced substantial size declines in recent years. The extent to which its force has shrunk is almost staggering, with a 22% decline in fleet size since 2006, and the lowest personnel numbers since 1935, according to various government sources. Its own keystone document on Air Power Doctrine admits the RAF *“cannot sustain high-intensity operations alone”*, with the number of MOD civil servants exceeding the number of active-duty troops in the RAF and the Royal Navy.

Its gaps in key capability areas, such as Air Mobility and ISTAR are concerning to outside observers, with pilot training times now at a high of 7.1 years. These effects will no doubt affect the ability of the RAF to protect British strategic interests and provide favourable outcomes.

These cuts are almost all as a result of gradual budget cuts by the MOD, especially from the 2010 Strategic Defence and Security Review, with inflation-adjusted spending on the MOD stagnating since 1956, and spending as a proportion of GDP dropping from 7.5% in the 1950s to just over 2% today. This has resulted in aircraft shortages, increasing its reliance on NATO, and diminishing its strength as an independently capable fighting force. The subsequent lack of public confidence around the RAF now means that sites like the Daily Mail can post misleading stories around the supposed ‘failures’ of the RAF, with the confidence that these stories will be believed by their readers.

Aim 2

Surprisingly, the RAF is still similarly sized to many European nations, and in some respects, excel above them. The trends in fleet size discussed in Aim 1 are not specific to the RAF, as shown by the House of Commons Defence Committee, suggesting that many nations are repurposing funds previously sent to the military for other uses. Moreover, the ‘special relationship’ between the UK and the USA practically guarantees that any conflict involving the UK would also receive support from not just the USA, the world’s largest superpower, but also NATO, the multinational organisation responsible for a majority of global military spending.

Compared to Russia and China, the RAF even has a more favourable fleet composition, albeit a far smaller one. However, the RAF is still in a good position to support NATO and is in a position of substantial safety and security by being an integral part of the alliance. Of course, the alliance does find itself on somewhat shaky ground in the aftermath of the US presidential election, but it can almost certainly rely on its European partners for years to come.

I was not able to estimate the extent to which recent scandals had had on public opinion towards the RAF. However, it is clear to see that the results suggested younger people had an unfavourable view of the RAF. The recent scandals embroiling the RAF have only served to compromise its image as a force for global good, as shown by statistical analysis, and universal condemnation from the press, the government, and the general public. The extent to which this has shifted overall public opinion could be an area for further development of the project.

It should be said that much of the work the RAF is conducting nowadays is away from the public eye, such as in the cyberspace and space domains, therefore reducing public awareness, and therefore public support. It uses the most highly advanced equipment, and as a very high degree of technological capability.

Aim 3

Despite the changes to its overall size, the RAF is still a capable fighting force, especially while fighting as part of or alongside NATO. It has been involved in multiple deployments over the past 10 years, including operations and exercises. Despite finding itself in tricky operational situations, it has managed to adapt to the situation and find success and has also become especially adept at providing air policing to allied nations as well as counteracting threats in the cyberspace domain.

The future for the RAF certainly looks bright, with gaps in capability set to be mended, with new E-7 Wedgetails to regain AWACS capability, new certifications on the Atlas C-1, and an increasing budgetary flow (2.5% of GDP) potentially providing it with the opportunity to increase its fleet size or enhance its capabilities as a multi-domain force.

The most effective ways to mend the RAF's reputation appear to be a change in culture and attitude, as well as further education about the role of the RAF. Although there are concerns amongst young people about the benefits of a career in the RAF, it is clear there are a number of benefits from a forces or forces-aligned career. Clearly, the best way to mend the image of the RAF from a public perspective is with education.

Overall, the RAF, at least independently, could not be considered as a capable, truly independent fighting force, as a result of the lack of budget, lack of aircraft, lack of personnel, and a culture that occasionally rears its ugly and outdated head to the public. Certainly, it is good enough to complete the tasks it is currently required to achieve, namely Quick Reaction Alert, Air Policing, and humanitarian missions. However, it should be argued that an air force that can seemingly only do 'the bare minimum' should not be commended for this effort.

Of course, with the assistance of NATO to patch its capability and personnel gaps, it is able to operate as a capable and influential fighting force. With the introduction of new aircraft to its fleet, it remains to be seen whether the RAF will have to rely on NATO for all conflicts, or whether it could project power in a local, or even regional conflict fully independently.

The effort to transform the RAF into an independent fighting force again will likely be long, and will most likely encounter setbacks. However, it is now down to RAF leadership, the MOD, and Government ministers to decide whether transforming the RAF into an organisation that truly stands up for global good is a task worth the effort.

Bibliography

Note: Sources are alphabetised by last name. Where a source is taken from an organisation, the source is be arranged alphabetically by the organisation's name.

If multiple sources are authored by the same organisation/person, they are arranged by the article's age. If there is still a conflict, the sources are organised alphabetically by the title of the source.

Source	CRAAP
Allison, George. (2022). <i>Plans to buy more transport aircraft scrapped due to cost</i>. [Online]. UK Defence Journal. Last Updated: 1 December 2022. Available at: https://ukdefencejournal.org.uk/plans-to-buy-more-transport-aircraft-scrapped-due-to-cost/ [Accessed 17 December 2024].	Currency- This article was published in March 2024. Although the figures reported on in the article could have shifted between the date of publishing and now, the article is reporting on the most recently published figures from the MOD, so the source can still be considered current. Relevance- The article is reporting on the drop in RAF aircraft numbers, with the sole focus of the article being this observation. As a result, the article is highly relevant to my project. Accuracy- The article uses the most recent data on total RAF aircraft numbers from the MOD, meaning it is likely to be reliable. Authority- This source was written by George Allison, a cybersecurity expert. Although he likely could not be considered an authority on defence matters, the article is published on the UK Defence Journal, a reputable website for news relating to UK defence matters, meaning the article comes from a reliable source. Purpose- The article's purpose is to highlight the drop in RAF aircraft numbers in recent years. Although they may stand to increase viewership, and therefore profit, by speculating and using emotive language, the article appears to be a simple explanation of reported fact, meaning the likelihood of bias being present is unlikely.
Allison, George. (2024). <i>How many aircraft do the RAF operate?</i>. [Online]. UK Defence Journal. Last Updated: 30 March 2024. Available at: https://ukdefencejournal.org.uk/how-many-aircraft-do-the-raf-operate/ [Accessed 17 December 2024].	Currency- This article was published in March 2024, making it somewhat current. Additionally, as fleet size numbers are very unlikely to face substantial changes in short periods (due to the complex processes involving introducing/retiring aircraft types), and the fact that the article uses the most recent publicly available government data, the article can be considered extremely current. Relevance- The article's sole focus is on the number and type of aircraft that the RAF operate, which is essential to my project. Therefore, it is extremely relevant. Accuracy- This article uses data from the Ministry of Defence's 2023 armed forces equipment and formations publication, published March 2024. Since the data comes from official and recent government sources, the source can be considered to be very accurate. Authority- This source was written by George Allison, a cybersecurity expert. Although he likely could not be considered an authority on defence matters, the article is published on the UK Defence Journal, a reputable website for news relating to UK defence matters, meaning the article comes from a reliable source. Purpose- The article's purpose is to bring attention to the types and number of aircraft that the RAF operate. Since the article is a reporting of published government statistics, the likelihood of bias being present is negligible.

Allison, George. (2024). <i>RAF aircraft numbers drop by 22% in seven years</i>. [Online]. UK Defence Journal. Last Updated: 4 March 2024. Available at: https://ukdefencejournal.org.uk/raf-aircraft-numbers-drop-by-22-since-in-seven-years/ [Accessed 17 December 2024].	Currency- This article was published in March 2024, making it somewhat current. Additionally, as fleet size numbers are very unlikely to face substantial changes in short periods (due to the complex processes involving introducing/retiring aircraft types), and the fact that the article uses the most recent publicly available government data, the article can be considered extremely current. Relevance- The article's focus is on the drop in the number of aircraft that the RAF operate, which is essential to my project. Therefore, it is extremely relevant. Accuracy- This article uses data from the Ministry of Defence's 2023 armed forces equipment and formations publication, published March 2024. Since the data comes from official and recent government sources, the source can be considered to be very accurate. Authority- This source was written by George Allison, a cybersecurity expert. Although he likely could not be considered an authority on defence matters, the article is published on the UK Defence Journal, a reputable website for news relating to UK defence matters, meaning the article comes from a reliable source. Purpose- The article's purpose is to bring attention to the types and number of aircraft that the RAF operate. Since the article is a reporting of published government statistics, the likelihood of bias being present is negligible.
Arora, Seerat. (2024). <i>Largest Air Forces in the World</i>. [Online]. AviationCareerHub. Last Updated: 2 August 2024. Available at: https://www.aviationjobsearch.com/career-hub/articles/career-advice/general/largest-air-forces-in-the-world [Accessed 17 December 2024].	Currency- This article was published in August 2024, meaning it is very current. Additionally, as fleet size numbers are unlikely to face any great increase/decrease in short periods of time, the article can be considered highly current. Relevance- The article discusses the largest air forces by both personnel and aircraft numbers. This is an integral part of Aims 1 and 2, therefore making the article very relevant. Accuracy- Although the article does not list its sources, and some numbers given in the source contradict other sources, the numbers provided in this source are approximately the same as numbers given by other sources ($\pm 5\%$), and are corroborated by other sources. Therefore, the article can be considered somewhat accurate. Authority- The article comes from AviationCareerHub, a website dedicated to finding employment in the aviation industry. Although it is not an authoritative matter on military topics, its aviation-adjacent area of focus means that it holds some authority on air force matters. Purpose- The purpose of the article is to inform readers of the size of the world's largest air forces. Due to the fact that the source is simply reporting on numbers, with these numbers being corroborated from other sources, the likelihood of bias is minimal.
Baker, Sinead. (2023). <i>Most Powerful Air Forces in the World in 2023, Ranked</i>. [Online]. Business Insider. Last Updated: 15 July 2023. Available at:	Currency- This article was published in late 2023, meaning it is somewhat dated. Furthermore, given that more recent data is available, the article does suffer a lack of currency. However, this can be overlooked a relative change in the strength of an air force takes significant time to change, so the rankings provided in the article can still be considered somewhat current. Relevance- The article's sole focus is on the relative strength of some of the world's most powerful air forces. This makes it relevant to Aim 2 of the project, making it relevant, and therefore reliable.

https://www.businessinsider.com/most-powerful-air-forces-in-the-world-in-2023-ranked-2023-6 [Accessed 17 December 2024].	Accuracy- The article uses data from GlobalFirepower, a website dedicated to determining the relative strength of various nations in a number of categories. Given the website is unique in its role, and uses government sources, it can be considered to be accurate. Authority- This article is written by Business Insider, a Business-oriented online news platform. Although it does not have strong links with military journalism, it is still a well-known and trusted news source, and can therefore be considered an authority on matters. Purpose- The purpose of this article is to inform readers of the relative strength of the strongest air forces. As it is a qualitative analysis, the article could be prone to bias, however its transparency about its sources used and lack of speculative language and evidence suggest that the possibility of bias is minimal.
Baker, Sinead; Spirlet, Thibault. (2023). <i>Ranked: the World's Most Powerful Militaries in 2023</i>. [Online]. Business Insider. Last Updated: 18 December 2023. Available at: https://www.businessinsider.com/ranked-world-most-powerful-militaries-2023-firepower-us-china-russia [Accessed 17 December 2024].	Currency- This article was published in late 2023, meaning it is somewhat dated. Furthermore, given that more recent data is available, the article does suffer a lack of currency. However, this can be overlooked a relative change in the strength of a country's armed forces takes significant time to change, so the rankings provided in the article can still be considered somewhat current. Relevance- The article's sole focus is on the relative strength of some of the world's most powerful armed forces. This makes it relevant to Aim 2 of the project, making it relevant, and therefore reliable. Accuracy- The article uses data from GlobalFirepower ,a website dedicated to determining the relative strength of various nations in a number of categories. Given the website is unique in its role, and uses government sources, it can be considered to be accurate. Authority- This article is written by Business Insider, a Business-oriented online news platform. Although it does not have strong links with military journalism, it is still a well-known and trusted news source, and can therefore be considered an authority on matters. Purpose- The purpose of this article is to inform readers of the relative strength of the strongest armed forces. As it is a qualitative analysis, the article could be prone to bias, however its transparency about its sources used and lack of speculative language and evidence suggest that the possibility of bias is minimal.
BBC. (2019). <i>Strength of British military falls for ninth year</i>. [Online]. BBC News. Last Updated: 16 August 2019. Available at: https://www.bbc.co.uk/news/uk-49365599 [Accessed 17 December 2024].	Currency- This article was published in 2019, meaning it is highly out of date. However, since it is simply highlighting an overall trend that had occurred at the time, this lack of currency will not affect its overall reliability. Relevance- The article details the drop in the overall strength of the British Armed Forces, and gives some reasons for this decline. This is relevant to both Aims 1 and 2, making this source reliable. Accuracy- The article uses data from the Ministry of Defence and statements from MPs, using the most recent data at the time the article was published. This makes it an accurate, and therefore reliable source, although this accuracy has somewhat diminished due to its lack of currency. Authority- This article was published by BBC News, a division of the BBC, a public service broadcaster. BBC News is renowned for reporting on reliable and

	corroborated evidence and stories, so it can therefore be considered a reliable authority on the matter. Purpose- The purpose of the article is to bring attention to the drop in strength of the UK Armed Forces. Since the majority of the BBC's revenue comes from a license fee, it does not stand to profit from sensationalism, which therefore mitigates the risk of bias.
Beale, Jonathan; Crew, Jemma. (2023). <i>Red Arrows: Predatory behaviour widespread and normalised - RAF</i>. [Online]. BBC News. Last Updated: 1 November 2023. Available at: https://www.bbc.co.uk/news/uk-67287479 [Accessed 17 December 2024].	Currency- This source was published in 2023, making it current. Additionally, as the article is reporting on an event, its currency is unlikely to diminish its overall reliability, therefore making it a reliable source. Relevance- The article is reporting on the occurrences of predatory and toxic behaviour within the RAF Red Arrows. As this is an integral part of Aim 2, this source is therefore very relevant, and therefore reliable. Accuracy- The article uses evidence from an inquiry into the matter, as well as quotes from the RAF and the House of Commons Defence Committee. As this source is simply using published government sources, the source is likely to have a high degree of accuracy. Authority- This article was published by BBC News, a division of the BBC, a public service broadcaster. BBC News is renowned for reporting on reliable and corroborated evidence and stories, so it can therefore be considered a reliable authority on the matter. Purpose- The article was published to bring attention to the occurrences of unacceptable behaviour within the RAF Red Arrows. Since the majority of the BBC's revenue comes from a license fee, it does not stand to profit from sensationalism, which therefore mitigates the risk of bias.
Brown, Larisa. (2024). <i>Civil servants in MoD outnumber navy and RAF personnel</i>. [Online]. The Times. Last Updated: 29 October 2024. Available at: https://www.thetimes.com/uk/defence/article/civil-servants-in-mod-outnumber-navy-and-raf-personnel-nngtrnfq0 [Accessed 17 December 2024].	Currency- This article was published in October 2024, making it highly current. Since figures like personnel working for the armed forces and civil service are unlikely to substantially change, it is very likely to contain reliable data. Relevance- The article's main focus is directed towards noting that the number of civil servants in the MOD now outnumber the number of serving military personnel in both the RAF and Royal Navy. Although this does not directly address the number of personnel serving in the armed forces, it is a very useful point of interest to note in the project, therefore making it relevant. Accuracy- The article uses figures from the MOD, meaning the source uses reliable information to form its points. It also notes how the number of civil servants was obtained, where short-term employees, contractors and those individuals working for non-departmental public bodies were excluded from the overall count. Authority- This article comes from The Times, an established news source, which has a 'high' factuality rating, according to news aggregation website Ground News. Additionally, the article uses data from the MOD, meaning it can be considered highly credible and accurate. Purpose- The purpose of this article is to bring attention to the number of MOD-employed civil servants compared to the number of serving military personnel. This is somewhat aligned with my project, therefore making this source reliable. However, it could be argued that this article was published as a way to cause controversy or create negative emotions around the MOD, in order to attract more attention,

	therefore leaving it susceptible to bias. However, as the article is primarily reporting on published fact, this possibility is unlikely.
Chapman, Khalem. (2018). <i>Airborne Early Warning and Control: Detecting the Battlespace</i> . [Online]. Aerosociety. Last Updated: 18 September 2018. Available at: https://www.aerosociety.com/news/airborne-early-warning-and-control-detecting-the-battlespace/ [Accessed 17 December 2024].	Currency- This article was published in 2018, making it very out of date. However, since the article discusses the uses and benefits of AEW&C aircraft, this lack of currency will not affect the source's reliability, since this information is not liable to change. Relevance- This source discusses the use of AEW&C aircraft, which is very relevant to Aim 1. This therefore makes the source highly reliable. Accuracy- This source reports on both historical and industry-known information, therefore meaning this source uses a wide array of accurate sources, thus making the source itself reliable. Authority- This source was written by Khalem Chapman, a reporter for the Royal Aerospace Society (RAeS), the sole professional body dedicated to aerospace, aviation, and space communities. This therefore makes the RAeS an authoritative both, thus making the source reliable. Purpose- The purpose of this source is to bring attention to the benefits of AEW&C aircraft. Since the article is a reporting of their uses, there is very little opportunity for loose speculation or bias, especially so since the RAeS gains the majority of its funding through memberships and not advertisement revenue through increased viewership. Therefore, the source has little chance of bias.
Defence Statistics (Tri-Service). (2017). <i>Request size of the army, navy and air force from 1700 to 2016</i> . [Online]. assets.publishing.service.gov.uk. Last Updated: 28 April 2017. Available at: https://assets.publishing.service.gov.uk/media/5a81d66740f0b623026996e1/2017-04440.pdf [Accessed 17 December 2024].	Currency- This source contains data up to 2016, and was published in 2017, so it is unlikely to represent current-day numbers. However, it does give a good indication in the overall size trend of branches of the armed forces, especially by containing data as far back as the 1700s. Relevance- This source contains exact data about the number of personnel in each branch of the British armed forces, including the RAF, making it highly relevant to the project. Accuracy- This document uses sourced information for all data after 1900, making it likely to be highly reliable. It is unlikely to contain bias as it has been published in response to a Freedom of Information (FOI) request, which, by law, are required to be accurate. Authority- This source is from the Defence Statistics (Tri-Service) branch of the Ministry of Defence, a Government source, meaning it is very likely to contain reliable data. Purpose- The purpose of this source is to provide data on the number of personnel in each branch of the British armed forces in response to a FOI request, meaning its purpose is clearly defined and relevant to the project, as well as minimising the potential for any bias.
Haynes, Deborah. (2023). <i>Defence cuts will affect UK's ability to fight from the air - well into the 2030s, MPs warn</i> . [Online]. Sky	Currency- This source was published in 2023, making it current. Additionally, since the source is a reporting of quotes, and explicitly mentions the time period for which the trends described in the source affect, the source can be considered highly current. Relevance- This source describes the effect budget cuts may have on the capability of the armed forces, which is relevant to Aim 1. This makes it highly relevant to the project.

News. Last Updated: 10 September 2023. Available at: https://news.sky.com/story/defence-cuts-will-affect-uks-ability-to-fight-from-the-air-well-into-the-2030s-mps-warn-12958250 [Accessed 17 December 2024].	Accuracy- This article uses statements from the House of Commons Defence Committee, previous reporting from Sky News, and official statistics from the MOD. This makes the source highly accurate, due to the authoritative nature of the sources used. Authority- This article was written by Deborah Haynes, an award-winning journalist who has reported in Iraq, Afghanistan, and Libya, and is now serving as Sky News' security and defence editor. This makes her a very authoritative source, therefore making this source highly reliable. Purpose- The purpose of this source is to bring attention to how budget cuts may affect the capability of the RAF. Although it makes money through advertisements, it is widely renowned as a reliable source that avoids speculative reporting, making it a reliable source, unlikely to be affected by bias.
Haynes, Deborah. (2023). <i>Royal Air Force unlawfully discriminated against white male recruits in bid to boost diversity, inquiry finds</i>. [Online]. Sky News. Last Updated: 30 June 2023. Available at: https://news.sky.com/story/royal-air-force-unlawfully-discriminated-against-white-male-recruits-in-bid-to-boost-diversity-inquiry-finds-12911888 [Accessed 17 December 2024].	Currency- This source was published in 2023, making it current. Additionally, Since the source is a reporting of a public inquiry, which will not change with time, the article can be considered highly current, provided the caveat that the source is describes as specifically implicating the RAF in 2023. Relevance- This source describes the unlawful behaviour the RAF was responsible for, which is highly relevant to Aim 2. This therefore makes the source highly relevant to the project. Accuracy- This source uses evidence from industry experts, statements from the RAF, and evidence from the inquiry into the matter. These sources are all highly reliable in their own right, therefore making this source highly accurate. Authority- This article was written by Deborah Haynes, an award-winning journalist who has reported in Iraq, Afghanistan, and Libya, and is now serving as Sky News' security and defence editor. This makes her a very authoritative source, therefore making this source highly reliable. Purpose- The purpose of this source is to bring attention to how budget cuts may affect the capability of the RAF. Although it makes money through advertisements, it is widely renowned as a reliable source that avoids speculative reporting, making it a reliable source, unlikely to be affected by bias.
Haynes, Deborah. (2023). <i>UK will be left 'dangerously exposed' when it cuts fleet of special forces aircraft, sources say</i>. [Online]. Sky News. Last Updated: 15 May 2023. Available at: https://news.sky.com/story/uk-will-be-left-dangerously-exposed-when-it-cuts-fleet-of-	Currency- This source was published in 2023, making it current. Additionally, as it is reporting on public statements, which will not change with time, the source can be considered current. Due to the fact that the RAF are still without special forces capability, the source is therefore very current, and therefore very reliable. Relevance- This source describes the effect cutting the C-130 fleet may have on the RAF's capability, which is very relevant to Aim 1. This therefore makes the source highly relevant to the project. Accuracy- This source uses evidence from industry experts and evidence from government sources. These sources are all highly reliable in their own right, therefore making this source highly accurate. Authority- This article was written by Deborah Haynes, an award-winning journalist who has reported in Iraq, Afghanistan, and Libya, and is now serving as Sky News'

special-forces-aircraft-sources-say-12881389 [Accessed 17 December 2024].	security and defence editor. This makes her a very authoritative source, therefore making this source highly reliable. Purpose- The purpose of this source is to bring attention to how budget cuts may affect the capability of the RAF. Although it makes money through advertisements, it is widely renowned as a reliable source that avoids speculative reporting, making it a reliable source, unlikely to be affected by bias.
Hill, John. (2024). <i>OCCAR says it resolved longstanding issues over A400M engine</i>. [Online]. Airforce Technology. Last Updated: 25 April 2024. Available at: https://www.airforce-technology.com/news/occar-says-it-has-resolved-longstanding-issues-over-a400m-engine [Accessed 17 December 2024].	Currency- This source was published in 2024, making it current. Additionally, since the source is a description of issues that have occurred, the source is very unlikely to be affected by a lack of currency, especially since it was published under one year ago. Relevance- This article describes the issues affecting the Airbus A400M (known as Atlas C-1 in the RAF), which is relevant to Aim 1 of my project. Therefore, the source is highly relevant, and therefore reliable. Accuracy- The source uses data and quotes from the MOD, House of Commons Defence Committee, and Airbus. Therefore, the source is very likely to be accurate, due to the quality of these sources of information this source is based on. Authority- This article was written by John Hill, a reporter for Global Defence Technology. Although both the site and journalist are not commonly known, Airforce Technology relies upon a network of over 40 business-to-business sites, which all independently acquire data from various reliable sources. Therefore, this source could be considered reliable. Purpose- The purpose of the source is to describe the issues affecting the engines of the A400M and how these issues are being mitigated. Since the article is reporting on data from the MOD, House of Commons Defence Committee, and Airbus, the likelihood of any bias is very low.
HM Government. (2024). <i>Defending Britain: leading in a more dangerous world</i>. [Online]. assets.publishing.service.gov.uk. Last Updated: 23 April 2024. Available at: https://assets.publishing.service.gov.uk/media/671b5d4e1898d9be93f75dc8/2024-04-23_Defending_Britain_-_FINAL.pdf [Accessed 17 December 2024].	Currency- This source was published in 2024, making it current. Since it is reporting a change in budgetary policy, it is unlikely to be affected by a lack of currency, provided the caveat that this change in budget is described as coming from 2024. Relevance- This source focuses on the planned change to the MOD's budget, which is relevant to Aim 3. This therefore makes it a relevant source to the project. Accuracy- This source is a report into a planned change into the funding of the MOD. Since this is purely a factual matter, which no opportunity for speculation, this source can therefore be considered accurate. Authority- This source comes from HM Government. Since this is a government source, reporting on a government matter, this source can be considered to come from an authority, and is therefore reliable. Purpose- The purpose of this source is to bring attention to the increase in overall funding to the MOD. Since this source is a government report, centred around a government decision, the likelihood of any bias being present is exceptionally low.
House of Commons Defence Committee. (2023). <i>Aviation Procurement</i>:	Currency- This source was published in 2023, making it current. Additionally, since it is publishing an analysis into the current state of the RAF, it uses long term trends and therefore makes long-term forecasts, which are likely to remain reliable for long periods.

<i>Winging it?</i>. [Online]. committees.parliament.uk. Last Updated: 5 September 2023. Available at: https://committees.parliament.uk/publications/41355/documents/203534/default/ [Accessed 17 December 2024].	Relevance- This source addresses many issues the RAF face today, including hold-ups in pilot training and fleet shortages, which is extremely relevant to all 3 aims. Therefore, this source is highly relevant. Accuracy- This source is a parliamentary white paper that uses data from the MOD. As white papers must have exceptionally high levels of accuracy, and the data is from a government ministry, the source can be considered to be very accurate. Authority- This source originates from the House of Commons Defence Committee, the parliamentary scrutiny for defence affairs. This source therefore originates from an authoritative source. Purpose- The purpose of this source is to highlight the current issues facing the RAF. Since this is a report commissioned by the House of Commons Defence Committee, the report is almost certain to contain reliable information, as it allows for MPs to advocate for the most favourable strategic decision. As such, the possibility of bias being present is negligible.
Kirk-Wade, Esme. (2024). <i>UK Defence personnel statistics</i>. [Online]. The House of Commons Library. Last Updated: 13 August 2024. Available at: https://commonslibrary.parliament.uk/research-briefings/cbp-7930/ [Accessed 17 December 2024].	Currency- These statistics are dated to 1 April 2024, making them highly current. Since the overall size of the armed forces will change gradually, it is very reasonable to say that the current size of the RAF reflects this figure closely. Relevance- This source addresses numerical trends in the overall size and changes in size of each branch of the armed forces. This is very relevant to the requirements of my project, specifically Aim 1 Objective 1, seeking the current size of the RAF. Accuracy- This source uses data from the Ministry of Defence, a government department. The data within the source is therefore very likely to be reliable. Authority- These statistics come from the House of Commons Library. These figures are therefore very likely to be reliable as they come from a service dedicated to allowing MPs to inspect and scrutinise legislation and current government activities. Purpose- The purpose of the source is to provide statistics relating to the size of the British armed forces, including but not limited to the overall size of each branch, joining and leaving trends, and the diversity of the British armed forces. As the article's purpose is closely related to the needs of my project, it is therefore highly reliable. Additionally, as this article is simply reporting numerical statistics, there is unlikely to be any bias that skews the picture these trends portray.
Kirk-Wade, Esme. (2024). <i>UK defence spending</i>. [Online]. House of Commons Library. Last Updated: 4 December 2024. Available at: https://researchbriefings.files.parliament.uk/documents/CBP-8175/CBP-8175.pdf [Accessed 17 December 2024].	Currency- This source was published in 2024, making it current. Additionally, since it is reporting on long-term trends in defence spending, it is unlikely to be affected by currency, especially since it uses long-term forecasts to estimate how the budget of the armed forces may change. Relevance- This source addresses changes in MOD budget, which is relevant to Aim 1. This therefore makes the source highly relevant to the project. Accuracy- The source uses government information, as well as statistics from the MOD, making the source highly accurate as a result. Authority- This source was authored by Esme Kirk-Wade, for the House of Commons Library. This is a highly accurate parliamentary source, therefore making the source originate from a highly authoritative organisation.

	Purpose- The purpose of this source is to bring attention to the changes to the MOD's budget. Since this is a parliamentary report using government data, there is very little need for speculation or bias, therefore making this source's purpose purely for the delivery of information.
Kuenssberg, Laura. (2024). <i>Return of unpredictable president puts UK defence spending top of agenda</i>. [Online]. BBC News. Last Updated: 9 November 2024. Available at: https://www.bbc.co.uk/news/articles/c9vnd17v2vmo [Accessed 17 December 2024].	Currency- This source was published in 2024, making it very current. Since this article is a prediction of what <i>may</i> occur due to a second Trump administration, this article is very current. Relevance- This source is reporting on how a Trump presidency may affect national and global security. Since this is a part of Aim 2, this makes this source very relevant to the project. Accuracy- This article uses previously reported fact from Trump's first administration, as well as statements from industry experts and sources close to the President-elect. This therefore means the article is accurate as a result. Authority- This source was written by Laura Kuenssberg, a renowned journalist and the BBC's flagship political interviewer. Additionally the source was published by BBC News, a renowned organisation known for publishing reliable information. Purpose- The purpose of this article is to bring attention to the effects that a second Trump presidency may have on national and global security and stability. Since the majority of the BBC's revenue comes from a license fee, it does not stand to profit from sensationalism, which therefore mitigates the risk of bias.
Lake, Jon. (2025). <i>Red Arrows scare-mongering</i>. [Online]. Aerospace Global News. Last Updated: 4 January 2025. Available at: https://aerospacglobalnews.com/news/red-arrows-scare-mongering/ [Accessed 6 January 2025].	Currency- This source was published in 2025, making it extremely current. Since it is reporting on the pitfalls of another article, it is very unlikely that a lack of currency will affect its reliability. Therefore, it could be considered extremely current. Relevance- The article focuses on how the Daily Mail told incorrect information in their report about the Red Arrows. This is relevant to Aim 1, therefore making this source relevant to the project. Accuracy- This source uses statements from the MOD, and directly uses quotes from the Daily Mail article, thereby making the article highly accurate. Authority- This source was written by Jon Lake, a well known published aviation writer who has contributed to a variety of publications. This therefore makes him an authority on aerospace matters. Purpose- The purpose of this article is to bring attention to how the Daily Mail's story into the Red Arrows was inaccurate and speculative.
Lima Charlie. (2023). <i>Lima Charlie: Why was the Hercules so valuable and what will we do without it?</i>. [Online]. Forces News. Last Updated: 14 June 2023. Available at: https://www.forcesnews.com/technology	Currency- This article was published in June 2023, meaning it is lacking currency. However, since this is an analysis of the C-130 Hercules' uses in the RAF and speculation on how it will be replaced, this lack of currency will have very little, if any, impact on its reliability. Relevance- This article focuses on the uses and possible alternatives to the C-130 Hercules within the RAF. This information is very relevant to the project, as it allows for the analysis of the RAF's current capability, particularly with regards to the Air Mobility fleet, after the retirement of the C-130. Accuracy- This article uses historical examples to explain the uses and capabilities of the C-130, both in service with the RAF as well as other air forces, such as the United

gy/aircraft/lima-charlie-why-was-hercules-so-valuable-and-what-will-we-do-without-it [Accessed 17 December 2024].	States Air Force. It also uses government data to explain the decisions behind its retirement, and the capability of the RAF to replace it. Authority- This article was written by 'Lima Charlie', a pseudonym for the contributors of Forces News to give their true and professional opinions under one editorial voice. Forces News is also an established news source on news relating to the UK's armed forces, making the article come from a strong source of authority. Purpose- This article was written to explain the importance of the C-130 to the RAF, for those who may not be as aware of its importance in the air mobility fleet. Although it could be subject to bias by invoking negative emotions in the readers in order to increase viewership, it comes to similar conclusions as many other publications, and details out the exact areas where the RAF will have issues, for example through the lack of special forces capability on the A400. As such, the likelihood of bias influencing the conclusions of the source is low, making it a reliable source.
Martin, Tim. (2023). <i>Airbus talks A400M airlifter risks, sees military order slump from Q1 results</i>. [Online]. Breaking Defence. Last Updated: 4 May 2023. Available at: http://breakingdefence.com/2023/05/airbus-talks-a400m-airlifter-risks-sees-military-order-slump-from-q1-results/ [Accessed 17 December 2024].	Currency- This source was published in 2023, making it very current. Since it is reporting on quotations and official statistics, a lack in currency is also unlikely to affect its reliability. Relevance- This source discusses the issues that have faced the A400M in recent years, which is relevant to Aim 1. This therefore makes the source very relevant. Accuracy- This source uses previously reported and corroborated fact, quotes, and official statistics, which makes the source highly accurate. Authority- This source was authored by Tim Martin, a twice-Defence Media Award recipient. This therefore makes him an authority on the topic. Purpose- The purpose of this source is to bring attention to the issues that have affected the A400M, the ways in which these issues have affected various air forces, and the measures taken by Airbus as a result. Since this article is a reporting of already established fact, it is unlikely there is bias present in this article, with very little opportunity for speculation in the article.
Ministry of Defence. (2022). <i>Joint Defence Publication O-30</i>. [Online]. assets.publishing.service.gov.uk. Last Updated: September 2022. Available at: https://assets.publishing.service.gov.uk/media/636baad0d3bf7f1649c4e36d/UK_Air_Power_JDP_O_30.pdf [Accessed 17 December 2024].	Currency- This source was published in 2022, making it current. Additionally, since this is the most up-to-date version of JDP O-30 published, this source is as current as possible. Relevance- This source describes the required capability of an air force to be successful, which is integral to all aims of the project. Therefore, this source is integral to the completion of the project. Accuracy- This source is a public release of basic UK air power doctrine. Since this is the defining source of this doctrine, it is therefore extremely accurate. Authority- This source was published by the MOD, the defining authority on UK defence matters. As such, this source comes from a highly authoritative organisation. Purpose- The purpose of the source is to allow readers to understand the basics of UK air power doctrine. Since this is a reporting of doctrine, there is no possibility of bias.

NATO Public Diplomacy Division. (2024). <i>Defence Expenditure of NATO Countries</i>. [Online]. North Atlantic Treaty Organisation. Last Updated: 12 June 2024. Available at: https://www.nato.int/nato_static_fl2014/assets/pdf/2024/6/pdf/240617-def-exp-2024-en.pdf [Accessed 17 December 2024].	Currency- This source was published in 2024, making it very current. Additionally, since military funding does not change substantially over short periods, the source can be considered highly current. Relevance- This source describes the military spending of all NATO member states, a key point of Aim 2. This therefore makes this source highly relevant. Accuracy- This source uses internal and government data to publish the provided figures. Since NATO is a multinational organisation dedicated to mutual defence, it can be considered to hold accurate data, therefore making the source accurate. Authority- This source comes from NATO, a multinational organisation dedicated to mutual defence. It can therefore be considered an extremely authoritative, and therefore reliable, source of information. Purpose- The purpose of this source is to bring attention to the amount of military spending by NATO member nation. As this source is a simple reporting of statistics, and NATO is a multinational organisation involving national governments, the likelihood of bias is exceptionally low.
Royal Air Force. (2019). <i>Royal Air Force Ethos, Core Values and Standards</i>. [Online]. RAF Recruitment. Last Updated: October 2019. Available at: https://recruitment.raf.mod.uk/recruitment/media/3897/20200703-raf_ap1_2019_rev_3_page_spreads.pdf [Accessed 17 December 2024].	Currency- This source was published in 2019, which means its currency may be limited. However, since this is the most recent internal RAF document relating to their expected standards, the source is highly current. Relevance- This source outlines the expectations of the RAF, which is relevant to Aim 2 of the project. Therefore, this source is highly relevant. Accuracy- Since this source is a self-reporting of expected internal standards, any information published in this document can be assumed to be accurate, therefore making the source highly reliable. Authority- This source was written and published by the RAF, the sole defining authority on its own internal standards. Therefore, this source originates from a place of authority. Purpose- The purpose of this source is to bring attention to the expected internal standards of the RAF. Since this is a self-reporting of expected behaviour, the possibility of any bias being present is exceptionally unlikely.
Royal Air Force. (2023). <i>Chief of the Air Staff Statement on RAF Recruiting Inquiry</i>. [Online]. Royal Air Force. Last Updated: 29 June 2023. Available at: https://www.raf.mod.uk/news/articles/chief-of-the-air-staff-statement-on-raf-recruiting-inquiry/ [Accessed 17 December 2024].	Currency- This source was published in 2023, making it current. Additionally, since this source is reporting on a statement that was made by the Chief of the Air Staff, its currency (or lack of) cannot affect its reliability. Relevance- This source was published as a public apology for the effects of the 2023 RAF recruitment scandal and subsequent inquiry into the matter. Since this is relevant to Aim 2, this makes the source relevant to my project. Accuracy- This source is a direct statement from the Chief of the Air Staff, Air Chf Mshl Sir Richard Knighton. As it is a direct statement, this source is extremely accurate. Authority- This source comes directly from the Royal Air Force, the UK's air and space force, and the organisation involved in the 2023 recruitment scandal. As such, they are the best authority on the matter of publishing this apology.

	Purpose- The purpose of this statement is to apologise to those affected by the 2023 recruitment scandal. Since the entire source is a direct quote, there is no possibility of bias.
Santos, Sofia Ferreira. (2024). <i>Trump ally says Ukraine focus must be peace, not territory</i>. [Online]. BBC News. Last Updated: 9 November 2024. Available at: https://www.bbc.co.uk/news/articles/czxrwr078v7o [Accessed 17 December 2024].	Currency- This source was published in 2024, making it very current. Additionally, the source is reporting on quotes and official statistics, which will not be affected by currency, given that they are reported in context of the time they are published, thus making this source highly current, and therefore reliable. Relevance- This source is reporting on how a Trump presidency may affect the defence of Ukraine against the ongoing Russian invasion. Since this is a part of Aim 2, this makes this source very relevant to the project. Accuracy- This source uses quotes from matters of authority, and official statistics and figures. Due to the publication of these figures, and the fact that the reported matters are corroborated by other sources, this makes this source very likely to be accurate. Authority- This article was published by BBC News, a division of the BBC, a public service broadcaster. BBC News is renowned for reporting on reliable and corroborated evidence and stories, so it can therefore be considered a reliable authority on the matter. Purpose- The article was published to bring attention to the effects of the imminent second presidency of Donald Trump may have on the defence of Ukraine. Since the majority of the BBC's revenue comes from a license fee, it does not stand to profit from sensationalism, which therefore mitigates the risk of bias.
Stockholm International Peace Research Institute. (2023). <i>The top 15 military spenders, 2022</i>. [Online]. Stockholm International Peace Research Institute. Last Updated: April 2023. Available at: https://www.sipri.org/visualizations/2023/top-15-military-spenders-2022 [Accessed 17 December 2024].	Currency- This source was published in 2023, making it current. Although it is reporting on official statistics, which are liable to change over time, it uses some of the most publicly available data, therefore making it reliable. Relevance- The source reports on the spending of different nations on their respective armed forces, which is an essential part of Aim 2. As a result, this makes the source very relevant to the project. Accuracy- This source uses self-acquired data, based on government releases and professional estimations. Although this may slightly compromise accuracy, SIPRI's experienced history of open source analysis makes it a reliable source. Authority- This source comes from SIPRI, an international institute dedicated to research into various military matters. As a platform for collaboration between international researchers, it can be considered an authoritative source. Purpose- The purpose of this source is to bring attention to the amount of military spending by country, Since a proportion of SIPRI's revenue originates from the Swedish government in the form of a grant, it is unlikely to rely on speculative or biased reporting for increased viewership, therefore increasing its overall reliability.
The World Directory of Modern Military Aircraft. (2024). <i>Global Air Powers Ranking</i>. [Online]. The World Directory of	Currency- This source was published in 2024, making it very current. Additionally, since this uses the most publicly available data the site is continuously updated, and relative military strength changes over long periods of time, the source can be considered very current. Relevance- This source is an analysis of the world's strongest air forces and their relative strengths. Since this is a key part of Aim 2, the source is therefore highly relevant.

Modern Military Aircraft. Last Updated: 2024. Available at: https://www.wdmdma.org/ranking.php [Accessed 17 December 2024].	Accuracy- This source uses the most up-to-date publicly available sources from various nations. This therefore makes the source accurate, although its exact methodology is not published on its website, compromising its reliability somewhat. Authority- This source comes from The World Directory of Modern Military Aircraft, the leading source for calculating the relative strength of various air forces. Therefore, the source has come from an authoritative matter, although it could be argued the organisation is not as authoritative as an established and more well-known media source. Purpose- The purpose of this article is to provide an estimation as to the relative strength of various armed forces. Since the site has the aim of being as accurate as possible, it is unlikely bias is present, especially since it discloses the statistics on which it bases its decisions.
Thomas, Richard. (2022). <i>UK airborne early warning capability gap widens</i>. [Online]. Airforce Technology. Last Updated: 29 November 2022. Available at: https://www.airforce-technology.com/news/uk-airborne-early-warning-capability-gap-widens/ [Accessed 17 December 2024].	Currency- This source was published in 2022, making it current. Additionally, as the UK is still lacking AEW&C capability, the source is therefore still highly current. Relevance- The source is reporting on the UK's current lack of AEW&C capability, which is an integral part of Aim 1. This therefore makes the source highly relevant to the project. Accuracy- The source reports on publicly available press releases, and quotations from MOD spokespersons. This therefore makes the sources highly likely to be accurate. Authority- This source was written by Richard Thomas, editor-in-chief for defence and transport at GlobalData. Due to his long tenure as a defence reporter (since 2014), he can be considered an authoritative, and therefore reliable, source of information. Purpose- The purpose of this article is to highlight the lack of AEW&C capability in the RAF's current fleet. Due to the fact that the article is reporting on publicly available data and quotations from industry sources, the likelihood of bias is low. Since the article is only being used in the project to highlight the current lack of AEW&C aircraft, this bias should not have an effect on any findings of the project.

Appendix

Analysis of Data Collection Methods

News Articles

Strengths:

- Timely and current information
- Wide reach and accessibility
- Often written by experts or journalists with access to primary sources

Weaknesses:

- Potential bias or sensationalism
- May not go into significant depth on complex issues
- May require subscription or access fees

Opportunities:

- Use of multimedia (videos, infographics) can help to enhance understanding of a topic
- Access to a variety of perspectives from different news outlets

Threats:

- Misinformation or fake news
- Rapidly changing news cycle can make information outdated quickly

Opinion Pieces

Strengths:

- Provides personal insights and perspectives
- Can stimulate discussion and debate
- Often written by thought leaders or experts

Weaknesses:

- Subjective and potentially biased
- May lack supporting evidence or data
- Can be polarizing

Opportunities:

- Can highlight emerging trends or issues

- Opportunity to engage with the author's viewpoints through comments or social media

Threats:

- Risk of echo chambers or confirmation bias
- Potential backlash or controversy
- Potential for bias depending on the author

Survey limited to those who are likely to know about the topic i.e. restricted access

Strengths:

- High relevance and accuracy of responses
- Can provide detailed and specific insights
- Easier to interpret and analyse data

Weaknesses:

- Limited sample size may affect conclusion of wider public views on a topic
- Potential for selection bias
- May require incentives to encourage participation

Opportunities:

- Use of targeted online platforms to reach knowledgeable respondents
- Ability to follow up with respondents for deeper insights

Threats:

- Risk of non-response or low participation rates
- Privacy concerns and data protection issues

Online Research Papers

Strengths:

- Access to peer-reviewed and credible sources
- Often includes detailed methodology and data
- Can cover a wide range of topics and disciplines

Weaknesses:

- May require access through academic institutions or subscriptions
- Can be highly technical and difficult to understand for non-experts
- Potential for publication bias

Opportunities:

- Increasing availability of open-access journals
- Use of citation networks to find related research

Threats:

- Risk of outdated information if not regularly updated
- Potential for retraction or correction of published papers

Government/Official Sources**Strengths:**

- Non-intrusive method of data collection
- Very likely to be accurate
- Requires little pre-emptive work to collect data (unlike a survey)
- Access to historical data

Weaknesses:

- Potential for data to be misinterpreted
- Limited to only what data has been official reported
- Data may be outdated, incomplete, or irrelevant

Opportunities:

- Can be cross-referenced with other data sources
- Allows for the use of digital archives

Threats:

- May be restrictions on what data is publicly available

General survey**Strengths:**

- Can reach a large audience quickly
- Data can be standardised and collated
- Requires little active effort to collect results

Weaknesses:

- Depth of responses could be limited
- Questions or answers may be misinterpreted by respondents or person conducting survey
- Potential for low response rate

Opportunities:

- Can be cross-referenced with other data sources
- Able to analyse large data sets for trends

Threats:

- Respondents may not take the survey seriously, requiring removal of bogus responses
- Data privacy concerns
- Respondents may be suffering from survey fatigue

Interview

Strengths:

- Can collect in-depth and detailed information
- Could build rapport with the interviewee
- Flexibility to spontaneously ask about new topics and adjust questions based on the direction of the interview

Weaknesses:

- Potential interviewer bias
- Questions asked may be unintentionally offensive or irrelevant
- Limited sample size
- Time-consuming and potentially costly

Opportunities:

- Could include questions from a survey, allowing for cross-referencing with other data sources
- Remote interviews are now easier to conduct
- Able to gather qualitative insight

Threats:

- Scheduling conflicts
- Interviewee may not be as knowledgeable on the topic as predicted

Survey:

Templates

...

Templates

● ● ●

 Templates

- ☐ It has a very good amount
- ☐ It has an acceptable amount
- ☐ It has an ok amount, but could probably have more
- ☐ It has far too few and urgently needs more
- ☐ It has too many
- ☐ I don't know/I am not sure

- ☐ I know a lot about this
- ☐ I know a bit about this
- ☐ I have heard of this, but don't know much about it
- ☐ I have never heard of this

☐ Under 15

☐ 15-16

☐ 17-18

☐ 19-35

☐ 36-60

☐ Over 60

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

The RAF is a significant force for good

Please enter at most 250 characters

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

The RAF is extremely capable

Templates

Templates

Templates

7. What was your overall opinion of the RAF BEFORE you heard of this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

8. What was your overall opinion of the RAF AFTER you heard of this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

9. In 2023, the Red Arrows came under significant controversy for predatory behaviour being widespread and normalised. How familiar with this are you? *

- ☐ I know a lot about this
- ☐ I know a fair amount about this
- ☐ I have heard of this, but don't know much about it
- ☐ I don't know anything about this

10. What was your overall opinion of the RAF BEFORE you heard about this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

11. What was your overall opinion of the RAF AFTER you heard about this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

12. Since 2016, it has been found that the RAF's overall size has dropped substantially, from 724 aircraft to 564. How familiar are you with this? *

- ☐ I know a lot about this
- ☐ I know a fair amount about this
- ☐ I am aware of this, but don't know much about it
- ☐ I don't know anything about this

13. What was your overall opinion of the RAF BEFORE you heard about this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

14. What was your overall opinion of the RAF AFTER you heard about this?

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

15. Has your opinion of the RAF changed after hearing about all of the controversies above? *

- ☐ Yes
- ☐ No

16. What is your opinion of the RAF after hearing about all of the controversies above? *

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Very negative opinion

Very positive opinion

Templates

- ☐ Under 2 months
- ☐ 2-6 months
- ☐ 6 months-1 year
- ☐ 1-2 years
- ☐ 3-5 years
- ☐ 5-10 years
- ☐ More than 10 years

Templates

	This method will not work at all	This method might work a bit	This method would be reasonably effective	This method would be exceptionally effective
Increasing the RAF's size	☐	☐	☐	☐
Increasing advertising and recruitment	☐	☐	☐	☐
Harsher punishments for people who promote a negative culture	☐	☐	☐	☐
Changing the RAF's leadership	☐	☐	☐	☐
Increase operational deployments	☐	☐	☐	☐
Reduce operational deployments	☐	☐	☐	☐

Please enter at most 250 characters

☐ Definitely

☐ Maybe

☐ Probably not

☐ Definitely not

☐ Not sure/Prefer not to say

Please enter at most 250 characters

Please enter an email

Submit

Transcript of Interview:

Note- All filler words have been omitted.

Sam Travaglia (00:00)

So, I'm here with Mr Ashment, he is the school OC (Officer Commanding) of the CCF (Combined Cadet Force), Sir, if you'd like to tell us about your other career path as well.

Mr Ashment (00:08)

Hi Sam, yeah so as well as being a music teacher and CCF lead in school, I am also part of the reserve army in a musical capacity with the Royal Welsh.

Sam Travaglia (00:22)

Brilliant, so let's jump into the questions. First question, why did you join the armed forces, and do you think other people can be attracted into the forces with that same reason?

Mr Ashment (00:30)

Well, I join the armed forces back in 2001, some time ago now, I did 8 years in the regular Army, like I said, I'm the reserves now- The reason for starting, for joining back in 2001 really was [as a] career path for me, as a keen musician and very limited musical posts in civilian workplace[s]. The Army was the path that I took, it meant that I could join at 16 and a half and still learn, so I studied with Open University, as well as being a professional musician and touring. [As for] the same reason you said 'could it be attraction to others' and yes, absolutely for the same reason today as it was back in 2001, whether it's Army, RAF, or Navy, they offer exactly the same career path now as they did back 24 years ago.

Sam Travaglia (01:30)

So, on that note, do you think that joining this year (2025 now), do you think that'll benefit one's career as much as it did when you joined?

Mr Ashment (01:37)

Yeah, yeah, exactly the same. I mean, there's possibly even more support now, because I know that [...] the forces, not just Army, but RAF and Navy, do a lot of graduate schemes now, in that, once you've completed your A levels, the military, the MOD, will sponsor certain students, they will pay for university courses, fees, and then the transition into forces for a certain amount of time, depending on the contract, as that graduate programme.

Sam Travaglia (02:08)

Okay, next question. So, since 2016, the RAF has faced a substantial drop in aircraft and personal numbers, which has put under some public pressure, as well as pressure from the government. Would you say this trend is a concern, or do you think it's just a sign we're living in safer time?

Mr Ashment (02:22)

No, it is a massive concern, in my opinion, you've mentioned [it's] putting it under public pressure. I think the forces have got a lot of political pressures, and the financial cuts and the decline in the financial support for the Ministry of Defence across-the-board, it is huge, and maybe another bit of research to see how much that funding has been cut, and as a result of that, why. You know, the decline in personnel, aircraft, resources we often hear about military personnel being sent into

operational zones now, with limited resources, which is a huge concern. And it's purely off that, in my opinion, purely off the back of huge financial strain that any of the MOD, forces are under.

Sam Travaglia (03:10)

So, my next question is, what do you think the reasons for this decline in fleet size is, would you say it's just budget, or would you say there's any other reasons, such as declining public support for the forces?

Mr Ashment (03:18)

No I think there's a huge part of financial struggle, which, you know, certain senior leaders within any of the forces are under, and for that reason, they've had to cut regiments, fleets, aircrafts, whether it's sky, land sea, warfare, it's a struggle. However, I think to keep up with the times, the forces may be a little bit behind the time in terms of the way that [...] the world is run at the moment, with digital focus, things like AI taking over [...] manpower jobs, and for that reason, [it's] possibly another reason why certain people, certain areas of [the] MOD, have been cut.

Sam Travaglia (04:05)

Okay, in 2023, the RAF Red Arrows came under a lot of controversy because of predatory behaviour that was widespread and normalised within the branch. Would you say this culture is limited to only the Red Arrows or extends further into the RAF, or even the entire MOD?

Mr Ashment (04:20)

I think beyond the entire MOD, I think in a workplace in general now, whether it's civilian or military, I think that the demands from workplaces puts people under particular pressure and stress. We know how impressive the Red Arrows are as a unit, and they have to be put under pressure because it's such a demanding post, isn't it, and it's such a risky post. So yes, I think their demands are tough, but probably for the right reasons, because if they slip at all, if they're not as focused as they need to be it, it could end catastrophically for them.

Sam Travaglia (05:07)

So a survey I conducted around the school found that less than half of people believe the RAF is a force for good, obviously, we know looking at the way they've involve themselves around the Gaza conflict as well, they've come under a lot of public scrutiny for that, would you agree with this sentiment, and what do you think this sentiment has culminated around the RAF?

Mr Ashment (05:26)

It's interesting, isn't it? And I think to get a wide range of results from a survey is really important. It depends on who you're asking- If you're asking a certain age group and they're not as knowledgeable about what the RAF can offer, perhaps that has a particular sway on the results. Whether that survey has been put to a wider audience and we're looking at age ranges or people in different workplaces, they might have different opinions, but it's interesting to think that the ones that you've asked as part of the survey have found that the forces is not [a force for] good. I'd probably not agree with that, purely for public safety, more than anything else. Behind the scenes, what each of these units are doing to protect the country and protect us in effect.

Sam Travaglia (06:24)

So, on the note, how effect did you think the armed forces are compared to when you joined? Obviously mentioned budgetary strains earlier on, do you think they still have the same power they had before, or do you think it's diminished quite a bit?

Mr Ashment (06:37)

I think they're still effective, and I think they're probably still as effective, but as we know, the security and the risks around the world now are probably more than when I joined in 2001, far less hand to hand combat within operation[s], far more now, cyber and digital attacks, and in terms of warfare, [it's] a lot now less face-to-face, and for that reason, I think, although I think we're still as effective, we have to be aware that there's a wider concern, around [...] the world.

Sam Travaglia (07:17)

Do you think a lot of the sentiment can kind of come from not seeing what the RAF is doing? There's a lot of work in the RAF done like the cyberspace and the space domains, do you think a lot this negative sentiment comes as a result of people not knowing what the RAF is focusing on now?

Mr Ashment (07:31)

Absolutely. Absolutely. I don't know whether the RAF in particular can be a little bit more public about what they can do, but at the same time, when we're talking about cyber and security threats, there is an element of security for themselves and you've can't- it's a breach of security, if we publicise things too much.

It's very difficult, isn't it? But I do agree and I think, when it in terms of publicity, maybe the RAF could advertise the fact that they are doing as much as what they're doing- Not just the RAF, but the MOD, they're doing what they're doing, again to protect public security safety.

Sam Travaglia (08:08)

So, on that note, how long do you think it will take the RAF to revive its public image?

Mr Ashment (08:14)

That's a difficult question. You know, it depends on if certain threats are limited, or certainly cut now, [the] RAF can focus on what they're doing to protect and to combat those threats. We know that these threats are going to widen. I've mentioned technology already, but as technology advances more, I think those threats are going to become more serious and wider concerns, and it's how military keep up with that threat, around the world. But at the same time, I still think it's really important to keep the public in the loop with what is going on and what the military are doing to protect us and to combat those threats.

Sam Travaglia (09:03)

If you were to place a very approximate timescale, would you put it in the domain of like months or years, to even decades?

Mr Ashment (09:10)

It would certainly be years, and I would like to think within single figures of years. So, between the next 2 to 5 years, really educate the public more about what is happening and what they're doing to protect, and then hopefully by year 10, as a longer term plan, where can they see themselves, and hopefully raising that public profile in a positive way.

Sam Travaglia (09:34)

On that note as well, what you think would be some of the most effective methods the RAF could take in order to successfully revive its public image?

Mr Ashment (09:41)

I think like most things, if you can start with kind of schooling and educating primary, secondary, college, university students, or people of that age, it kind of feeds through then into society,

educating those people, I think, is the heart of it, rather than starting with the older generations and then the message not filtering through as much. And, yeah, working within communities as well, and that's really important, so looking with local councils, local politicians, in order to get their support and for that to spring.

Sam Travaglia (10:18)

So do you suppose a lot of people my age would have grown up seeing obviously Afghanistan and Iraq on the news, they wouldn't have been very sure about what was happening there, but all they would have seen was destruction and that could have fed into that sentiment, especially with Operation Pitting as well, the evacuation of Kabul, that would have pretty fed into that.

Mr Ashment (10:35)

On the media and on that point, the media and who to believe? And you know, as well as anybody, the correct sources, correct information, the most believable information, there's a lot of information online and within the media that has been stretched, let's say, and maybe not as believable as it should be, so I think that, the media also need to get on board with the fact that they've got to support, and they've got to be as honest and truthful as they possibly can be as well, that would really help.

Sam Travaglia (11:08)

So, one final question, do you think the armed forces are still able to project power and protect British strategic interests?

Mr Ashment (11:15)

Yes, absolutely. Yeah, I firmly believe that- I mentioned budget and financial cuts. However, I genuinely believe that [the] MOD, Army, RAF, Navy, are still doing the best that they can do with limited resources, and yeah, UK security at the heart of why it's being done.

Sam Travaglia (11:40)

I just want to quickly circle back to question 5, you mentioned that the amount of predatory behaviour is widespread beyond the MOD. Would you say it's a particular problem anywhere within the MOD, or it's about the same as you'd find within public life as well?

Mr Ashment (11:54)

No, I think the military have done a lot to get on board with diversity, equality, and making sure that people are being treated fairly, to the point where if somebody is feeling stressed or under pressure because of the job that they're in, very much like in civilian society, the Army have learned from that, the RAF have learned from that, the Navy have learned from the civilian approach, and they've definitely included that, certainly within the last I'd say, probably [post-] COVID pandemic, much more awareness around '*How are you feeling?*', '*Are you able to cope?*', and the right avenues to seek support if you're not coping. So yeah, yeah.

Sam Travaglia (12:40)

I'm going to ask one last question, so obviously you're the OC of the school CCF, how would you think organisation like the CCF, and the ATC (Air Training Corps) as well can help promote the RAF's public image and help revive its public image?

Mr Ashment (12:51)

Hugely. Like I said, I think it was [my answer to] your question 10, around you know, '*What can the British do?*', '*What can the forces do to project power to kind of spread the word more?*', I think us

having a unit in school, there is no better way to spread that message of why we're doing what we're doing. The skills, the qualifications, the life experiences that cadets, military personnel can have whilst in them in any form of military and the good it can do for everybody: society, communities. Yeah, and that message is strong, isn't it?

Sam Travaglia (13:31)

Sir, thank you, you've been amazing helpful. I'm going to write them up and I'll get that sorted. Thank you very much.

Mr Ashment (13:36)

Thank you Sam.